LCA of a 100kW PEMFC Stack

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Introduction

Proton Exchange Membrane Fuel Cell (PEMFC)

Reactions involved:

Anode: $H_2(g) \longrightarrow 2H^+(aq) + 2e^-$

Cathode: $1/2\ O_2 + 2H^+(aq) + 2e^- \longrightarrow H_2O(l)$

Overall: $H_2(g) + 1/2O_2(g) \longrightarrow H_2O(l)$

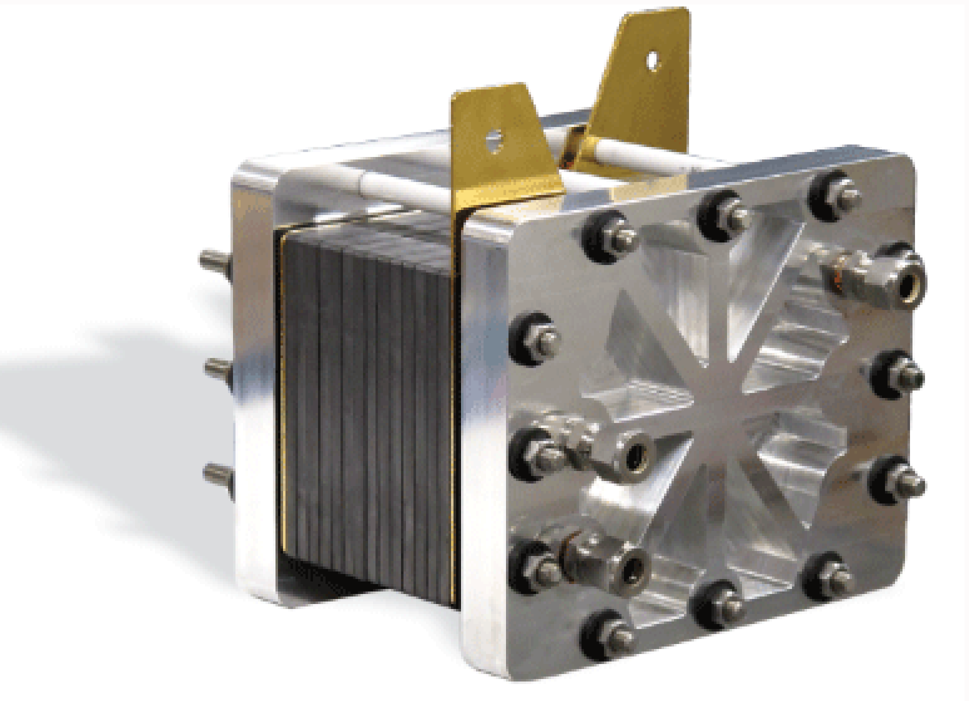


Figure 2: PEMFC Stack

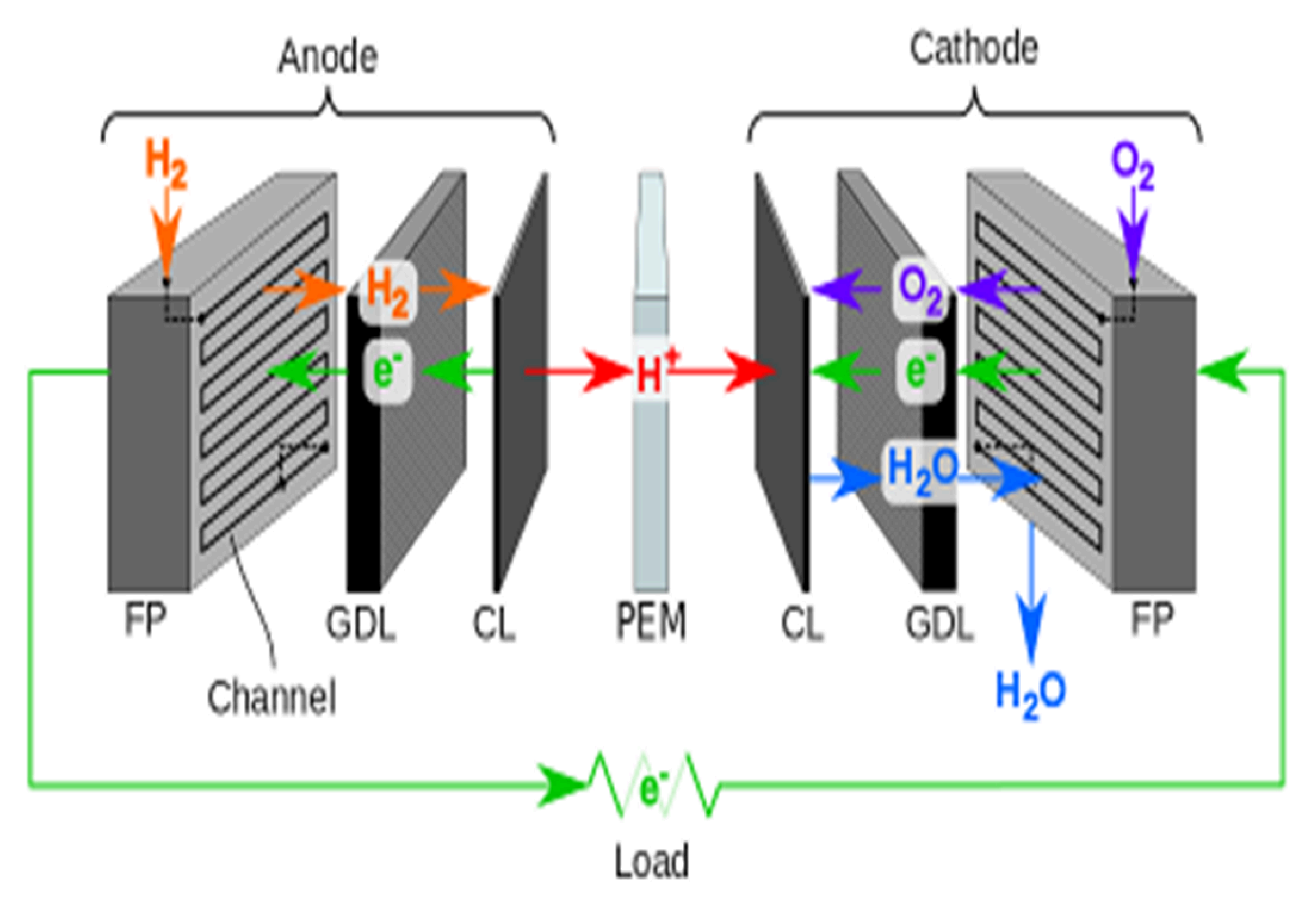


Figure 1: Schematic Diagram of PEM Fuel Cell

Goal of the LCA study

Study: Attributional LCA of a 100kW state-of-the-art PEMFC stack

Intended application: Intra-tier (modular structure) hotspot analysis of a State-of-the-art PEMFC system to identify potential main contributors for eco-design purposes.

Application status: Multiple studies like that of Riemer et al. (2023) provide a hotspot analysis of PEMFC stacks.

Gaps identified in existing literature:

- No harmonized structure
- Operation phase excluded
- Outdated design (majorly BPP coatings)
- Improper modelling/proxies used for Nafion.

Reasons for carrying out study: To serve as baseline for a comparative study against a Nafion free stack in the ECOPEM project.

Modelling approach: Cradle to grave LCA based on recycled content (cut-off) approach.

Target audience: Technical experts of the ECOPEM project

Scope of the LCA study

Functional unit: 1 kWh of net electricity delivered from a 100 kW Proton PEMFC stack operating within Europe during 2023.

Reference flows:

- Stack: 1.67x10^-6 units of a 100kW PEMFC stack (36 secs used out of its 6000h lifetime).
- Operation consumables: 0.0555 kg of hydrogen (5.55kg H2/hr at 2.5 bar) and 3.10kg of air (310.06 kg air/hr at 2.5 bar).
- Balance of Plant: $1.25x10^{-7}$ unit of BoP (36 secs used out of its 80000h lifetime).
- End of Life: Incineration of 1.38x10^-5 kg MEA and landfilling 5.63x10^-4 (rest of the stack) used.

Cut-off: The low-grade heat produced from the system is not valorized and cut-off.

Multi-functionality aspects: Water is classified as a waste, not a co-product due to its negligible economic value compared to the functional unit.

Impact assessment: The ImpactWorld+ 2.1, footprint version is used for impact assessment of the model built on Activity Browser (AB) version 2.11.2.

Limitations:

- Due to lack of data on recycling approaches, the MEA is incinerated while the rest of the stack is landfilled.
- A major component of the balance of plant, the humidifier is excluded from the model due to lack of data on its LCI.

Modular Structure

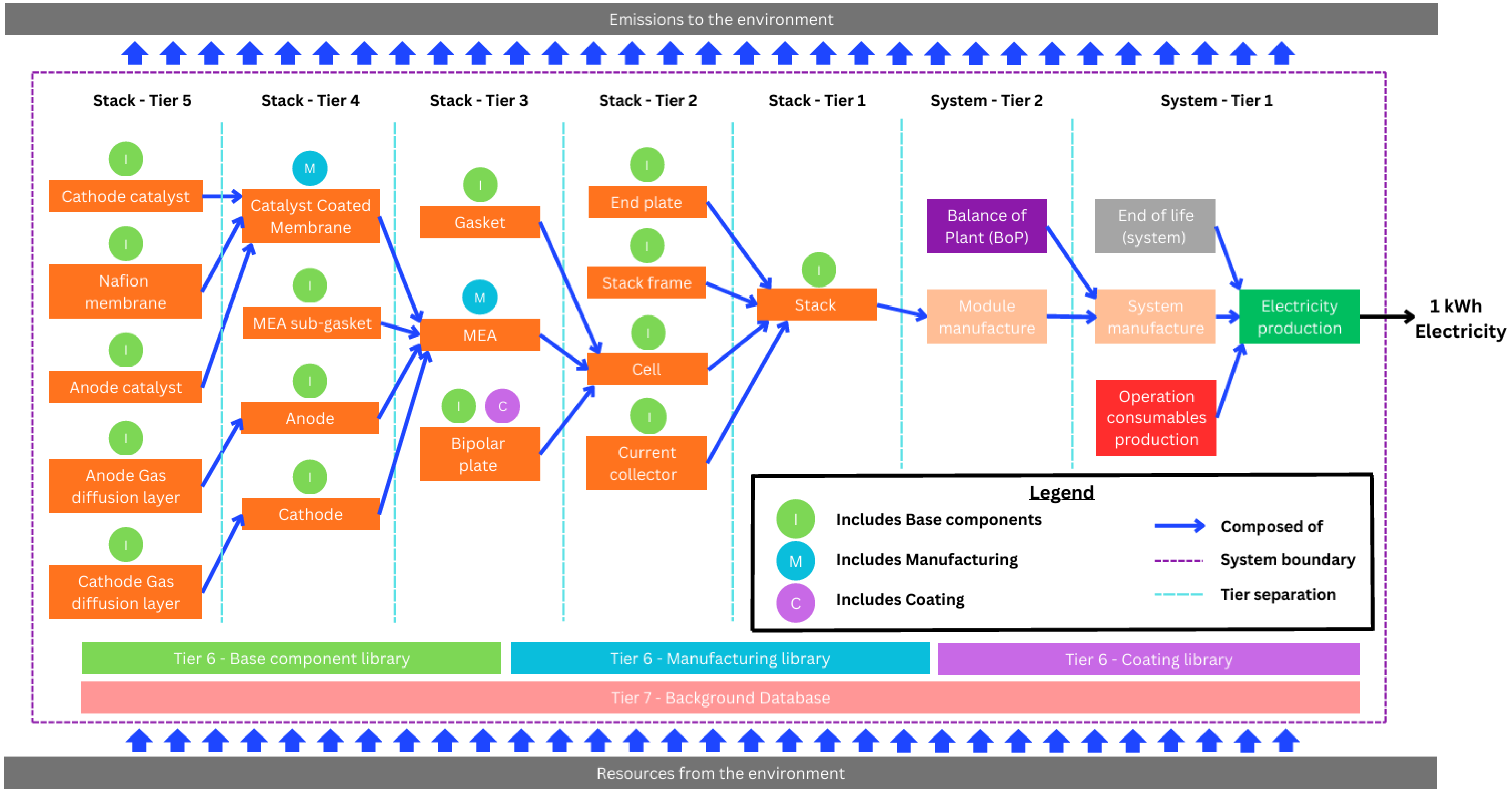


Figure 3

Modular Structure (Synchronized Technology)

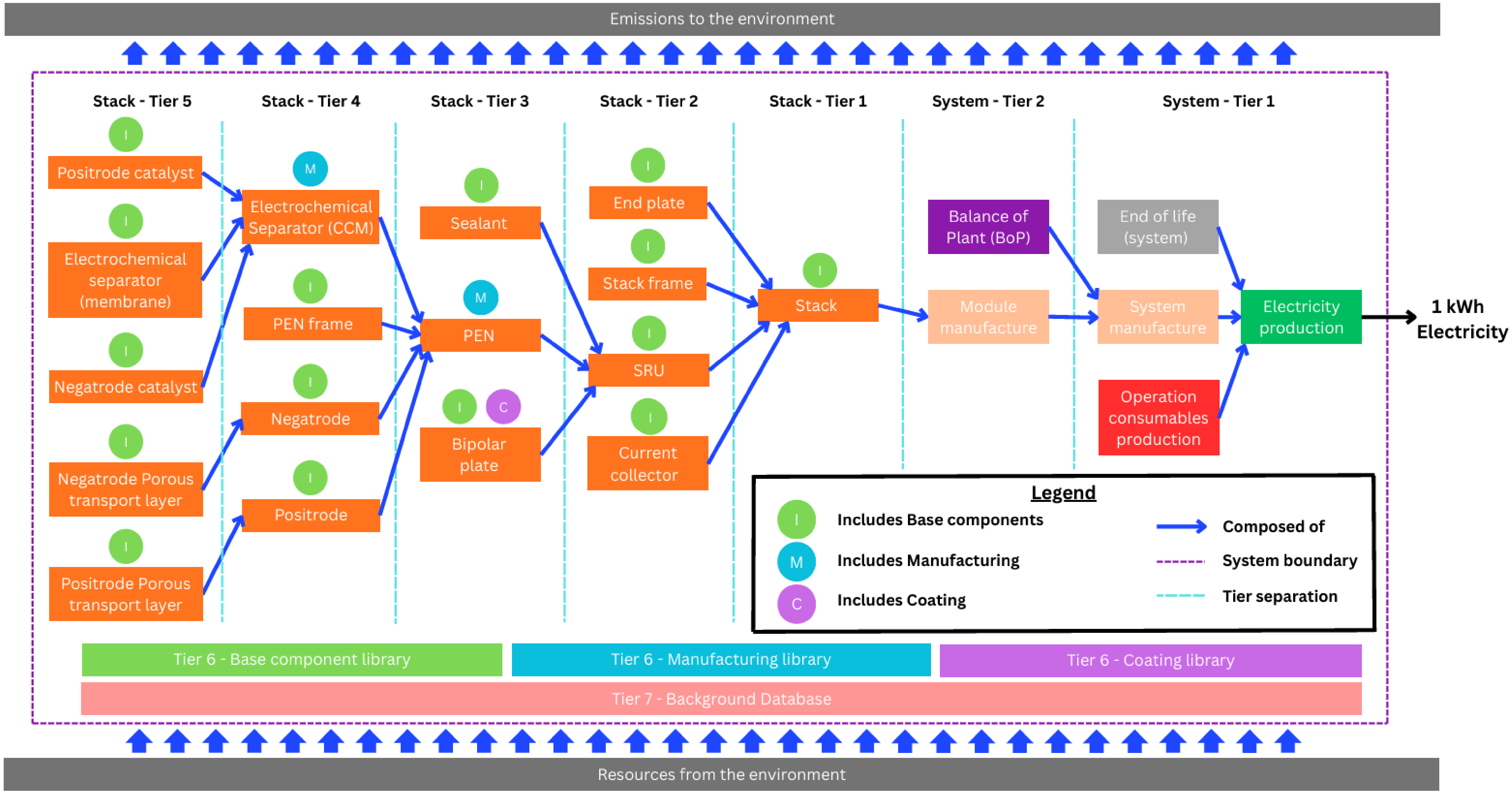


Figure 4

Life Cycle Inventory

- The parameters of the stack were derived from Riemer et al.(2023).
- The design of the stack was based on James et al.(2018).
- A sensitivity analysis was conducted to decide the LCI of the Nafion (membrane) used.
- Additions were made in the existing material database (HES-SO) to adress the gaps in the inventory of PEMFC stack.
- A state-of-the-art PVD based TiN coating was modelled to replace the outdated coating in most research papers.
- Evangelisti et al.(2017) and some other sources were used to fill the gaps and complete the ecoinvent-3.11-cutoff and biosphere3 based LCI in AB.

Insert quantities for each material or energy inputs specified in the process tree. For example in the frame manfacture, specify the weight of stainless steel required for the full frame. The column "Range" allows to specify how the input/output quantity can vary in the worst and best case (min/max). In output, include the main output (1 PEMFC stack) as well as potential co-products and potential wastes or emissions coming as an output of the	Source: Evangelisti et al. 2017 (as mentioned in Reimer), It is based on Battelle (2013) [Riemer's source]	Highlights in this colour in the column imply that the activity is to be found in the material database.

Technical specification of the product

Characteristic	Value	Unit	Data quality
Stack Power		KW or H2 MJ equivalent	
Total active area		m2	
Weight of the stack		kg	
Repetitive unit per stack		units	
Voltage		volts	
Degradation rate		%	

TABLE OF INPUT

Location of	Process	Type of flow	Flow/ Material name	Ecoinvent (or "material database")	Quantity	Range	Unit of	Data	Main features/notes/ assumptions
Europe without Switzerlan	Process 1: Nafion DE-521 solution manufacture	Auxiliaries	water, deionised	market for water, deionised	0.4499		kg	2	multiplied by (1/0.409) to scale the output to 1kg
GLO	Process 1: Nafion DE-521 solution manufacture	Raw materials	1-propanol	market for 1-propanol	0.5000		kg	2	
GLO	Process 1: Nafion DE-521 solution manufacture	Raw materials	nafion	Nafion production	0.0500		kg	2	
Europe without Switzerlan	Process 2: Cathode catalyst manufacture through ball milling	Auxiliaries	water, deionised	market for water, deionised	0.2941		kg	2	Scaled to 1kg output (multiplication factor = (1/0.13)
	Process 2: Cathode catalyst manufacture through ball milling	Raw materials	Nafion DE-521 solution		0.2059		kg		Qty based on Reimer (Refer "Other calculations").
RoW	Process 2: Cathode catalyst manufacture through ball milling	Raw materials	methanol	market for methanol	0.2941		kg	2	Scaled to 1kg output (multiplication factor = (1/0.13)
RER	Process 2: Cathode catalyst manufacture through ball milling	Energy consumptions	electricity, medium voltage	market group for electricity, medium vol	3762.4801		MJ	2	Scaled to 1kg output (multiplication factor = (1/0.13)
GLO	Process 2: Cathode catalyst manufacture through ball milling	Raw materials	platinum	market for platinum	0.0343		kg	2	Qty based on Reimer (Refer "Other calculations").
GLO	Process 2: Cathode catalyst manufacture through ball milling	Raw materials	carbon black	market for carbon black	0.1716		kg	2	Qty based on Reimer (Refer "Other calculations").
									Process 1 scaling factor:
									2.4450
									Process 2 scaling factor
									7.9491

Figure 5: A glimpse of the LCI data collection.

Nafion SA

Technosphere Flows:

Amount	Unit	Product	Activity	Location	Database	Formula
4e-10	unit	chemical factory, organics	market for chemical factory, organics	GLO	ecoinvent-3.11-...	
0.11	kilogram	soda ash, dense	market for soda ash, dense	GLO	ecoinvent-3.11-...	
0.6	kilogram	sodium hydroxide, witho...	market for sodium hydroxide, without water, in 50% solution state	RoW	ecoinvent-3.11-...	
3	kilogram	sodium hypochlorite, ...	market for sodium hypochlorite, without water, in 15% solution state	RoW	ecoinvent-3.11-...	
0.5	kilogram	sulfur trioxide	market for sulfur trioxide	RER	ecoinvent-3.11-...	
1.3	kilogram	tetrafluoroethylene	market for tetrafluoroethylene	GLO	ecoinvent-3.11-...	
5.22	ton kilometer	transport, freight, train, ...	market group for transport, freight, train, fleet average	RER	ecoinvent-3.11-...	
39.31	megajoule	heat, district or industrial, ...	market group for heat, district or industrial, natural gas	RER	ecoinvent-3.11-...	
0.87	ton kilometer	transport, freight, lorry, ...	market for transport, freight, lorry, unspecified	RER	ecoinvent-3.11-...	
-0.11	kilogram	waste plastic, mixture	treatment of waste plastic, mixture, municipal incineration FAE	CH	ecoinvent-3.11-...	
-0.38	kilogram	bilge oil	market for bilge oil	Europe without ...	ecoinvent-3.11-...	
-2.81	kilogram	spent solvent mixture	market for spent solvent mixture	Europe without ...	ecoinvent-3.11-...	
3.2	kg	Hexafluoropropene	Hexafluoropropene production	GLO	Material_db_v1(...	
-0.0848	kilogram	refinery sludge	treatment of refinery sludge, hazardous waste incineration	Europe without ...	ecoinvent-3.11-...	

Biosphere Flows:

Amount	Unit	Flow Name	Compartments	Database	Formula
0.0888	kilogram	Carbon dioxide, fossil	air	biosphere3	
2.36	kilogram	Sodium chloride	water	biosphere3	
1.87	kilogram	Sodium hydroxide	water	biosphere3	

Downstream Consumers:

Welcome Project: PEMFC ei v.3.11

Figure 6: Virgilio et al.(2023) LCI

S1 Inventory data for the production of PEMFC membranes

	2012	2020	Per kW of stack	Comments
Tetrafluoroethylene (TFE), at plant, EU	11.8	10.9	g/kW	57.4wt% TFE to 42.6wt% PSF.
Sulphuric acid, at plant, EU	8.8	8.1	g/kW	As a proxy for PSF. 42.6wt% PSF to 57.4wt% TFE.
Titanium dioxide, at plant, EU	n.a	1.9	g/kW	Hydrophilic additive, added in 2020 at a 10wt% ratio.
Transport, freight, rail, EU	0.0124	0.0125	tkm/kW[a]	Standard ecoinvent freight transport distances in Europe
Transport, freight, road, Euro4, EU	0.0021	n.a	tkm/kW	Standard ecoinvent freight transport distances in Europe
Transport, freight, road, Euro5, EU	n.a	0.0021	tkm/kW	Standard ecoinvent freight transport distances in Europe
Extrusion, plastic film, EU	21.0	21.5	g/kW	Used to represent the process of forming the polymer membrane film. Includes the energy demands (0.66 kWh_{el}/kg, 0.81 MJ_{th} /kg & 0.058 kg_{steam} /kg) and other consumables.

[a] tkm is the abbreviation of ton*km.

Figure 7: Simons et al.(2015) LCI

Table S9. LCI for the production process of a polymer electrolyte membrane.

Production of a polymer electrlyte membrane (Nafion)

Imputs	Amount unit	Remark
Tetrafluoroethylene, at plant/RER U	9.3E-01 kg	copolymer tetrafluorethylene; mol-ratio copolymer 1:copolymer 2 equals 7.5:1
Acrylonitrile-butadiene-styrene copolymer, ABS, at plant/RER U	3.1E-02 kg	
Styrene-acrylonitrile copolymer, SAN, at plant/RER U	3.1E-02 kg	represents copolymer 2: 3,6-dioxa-4-methyl-7-octene sulfonyl fluoride with equal shares for each copolymer, mol-ratio copolymer 1:copolymer 2 equals 7.5:1
Ethylene vinyl acetate copolymer, at plant/RER U	3.1E-02 kg	
[sulfonyl]urea-compounds, at regional storehouse/RER U	3.1E-02 kg	
Calandering PVC foil B250	1.0E+00 kg	energy consumption included in the process
Ammonia, liquid, at regional storehouse/RER U	9.7E-01 kg	used to "boil" the copolymers, value is an estimate
Ammonia, liquid, at regional storehouse/RER U	9.3E-03 kg	
Potassium chloride, as K2O, at regional storehouse/RER U	4.1E-02 kg	represents the production of potassium amide
Sodium, chloride electrolysis	1.3E-02 kg	
Heat, unspecific, in chemical plant/RER U	3.0E-02 MJ	heat required for the production of potassium amid
Electricity, low voltage, production UCTE, at grid/UCTE U	1.1E+00 kWh	electricity required for the production of potassium amid
Chemical plant, organics/RER/I U	4.0E-10 p	Standard from ecoinvent guidelines
Transport, freight, rail/RER U	8.3E-01 tkm	Standard distances from ecoinvent guidelines
Transport, lorry >16t, fleet average/RER U	2.1E-01 tkm	Standard distances from ecoinvent guidelines
Outputs		
Polymer electrolyte membrane, at plant	1.0E+00 kg	Final product
Heat, waste	3.8E+00 MJ	From electricity input
Hydrogen	5.5E-04 kg	from the reaction of liquid ammonia with potassium chloride
Disposal, municipal solid waste, 22.9% water, to municipal incineration/CH S	3.2E-02 kg	NaCl from the production of potassium amide

Figure 8: Schropp et al.(2024) LCI

Climate change, carbon footprint

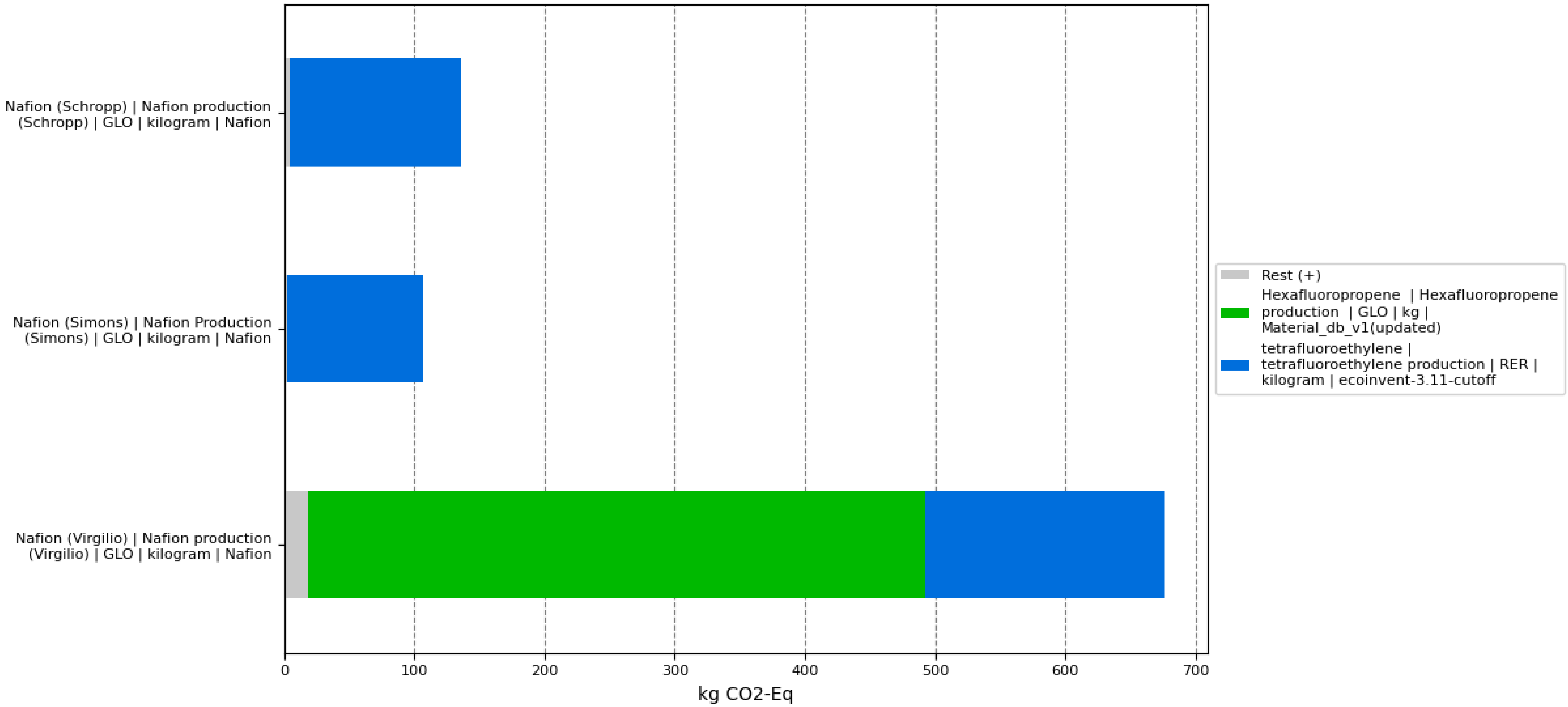


Figure 9: Reference flow: 1kg Nafion production

Ecosystem quality, remaining ecosystem quality damage

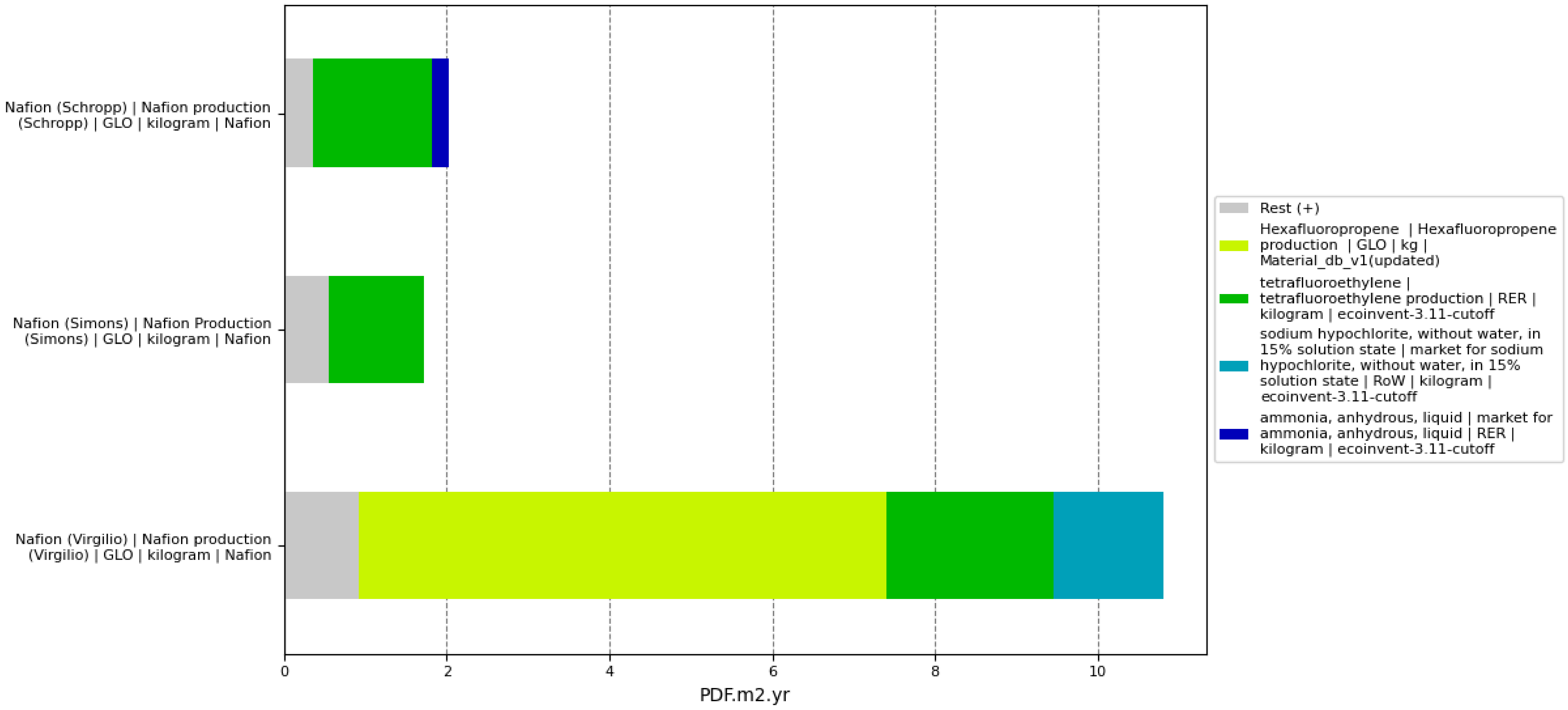


Figure 10: Reference flow: 1kg Nafion production

Energy resources: non-renewable, fossil and nuclear energy use

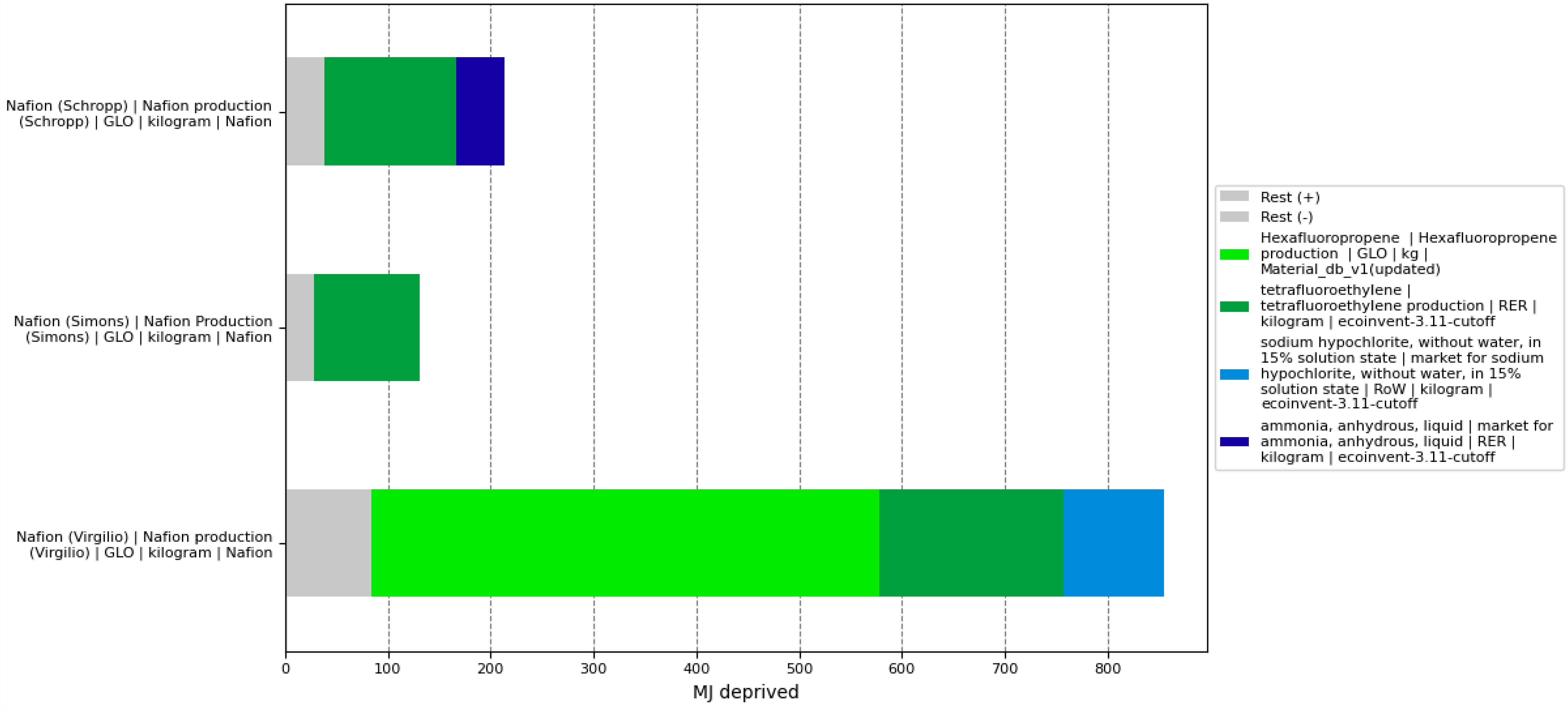


Figure 11: Reference flow: 1kg Nafion production

Human health, remaining human health damage

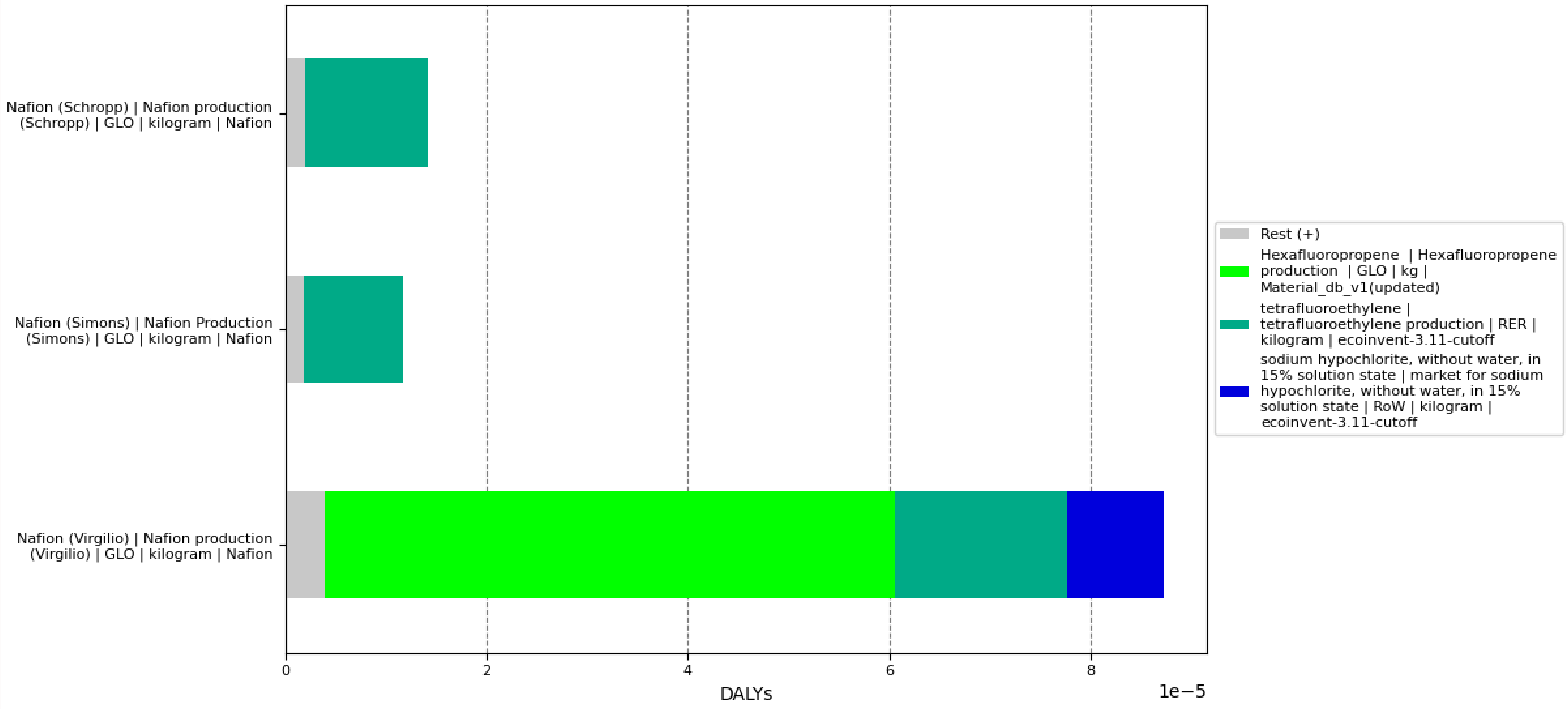


Figure 12: Reference flow: 1kg Nafion production

Water use, water scarcity footprint

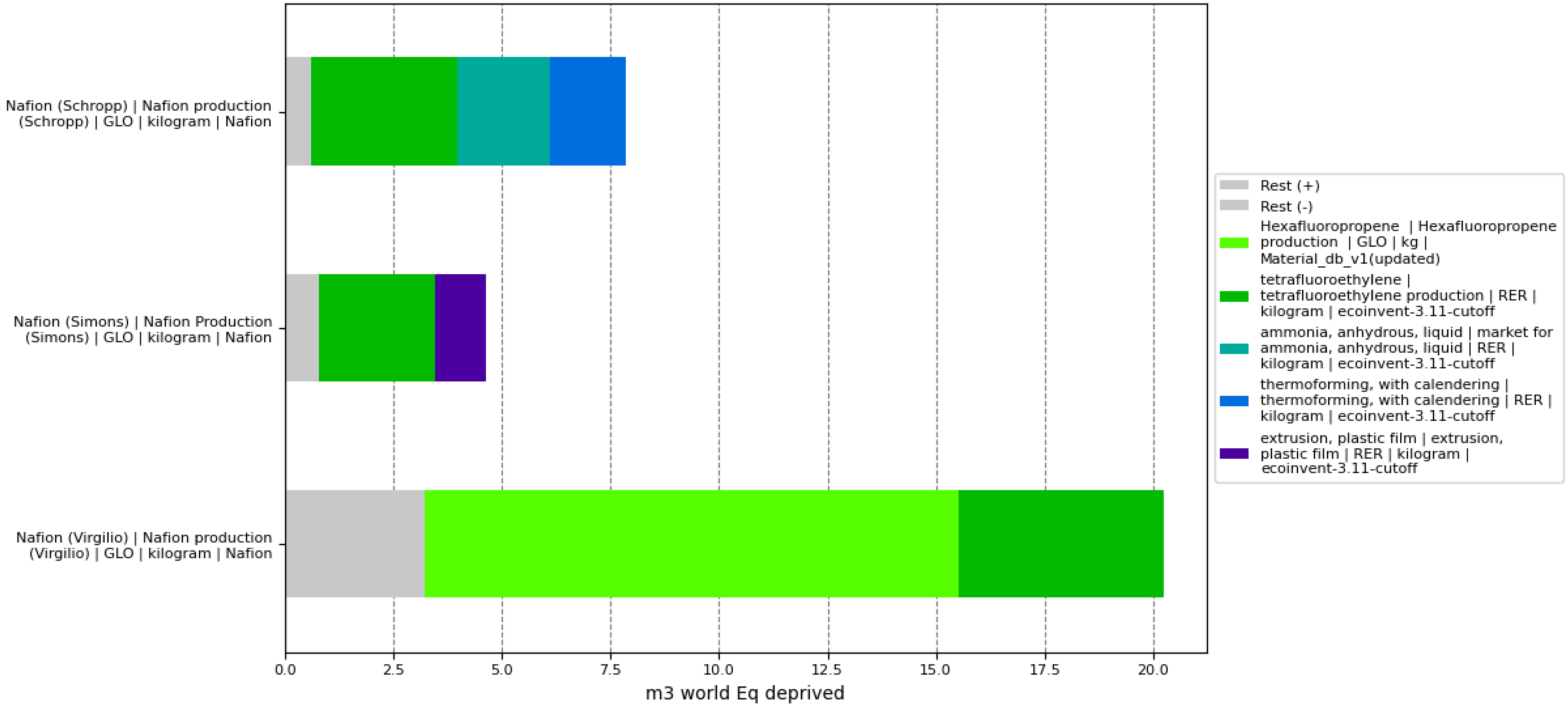


Figure 13: Reference flow: 1kg Nafion production

Sankey diagrams

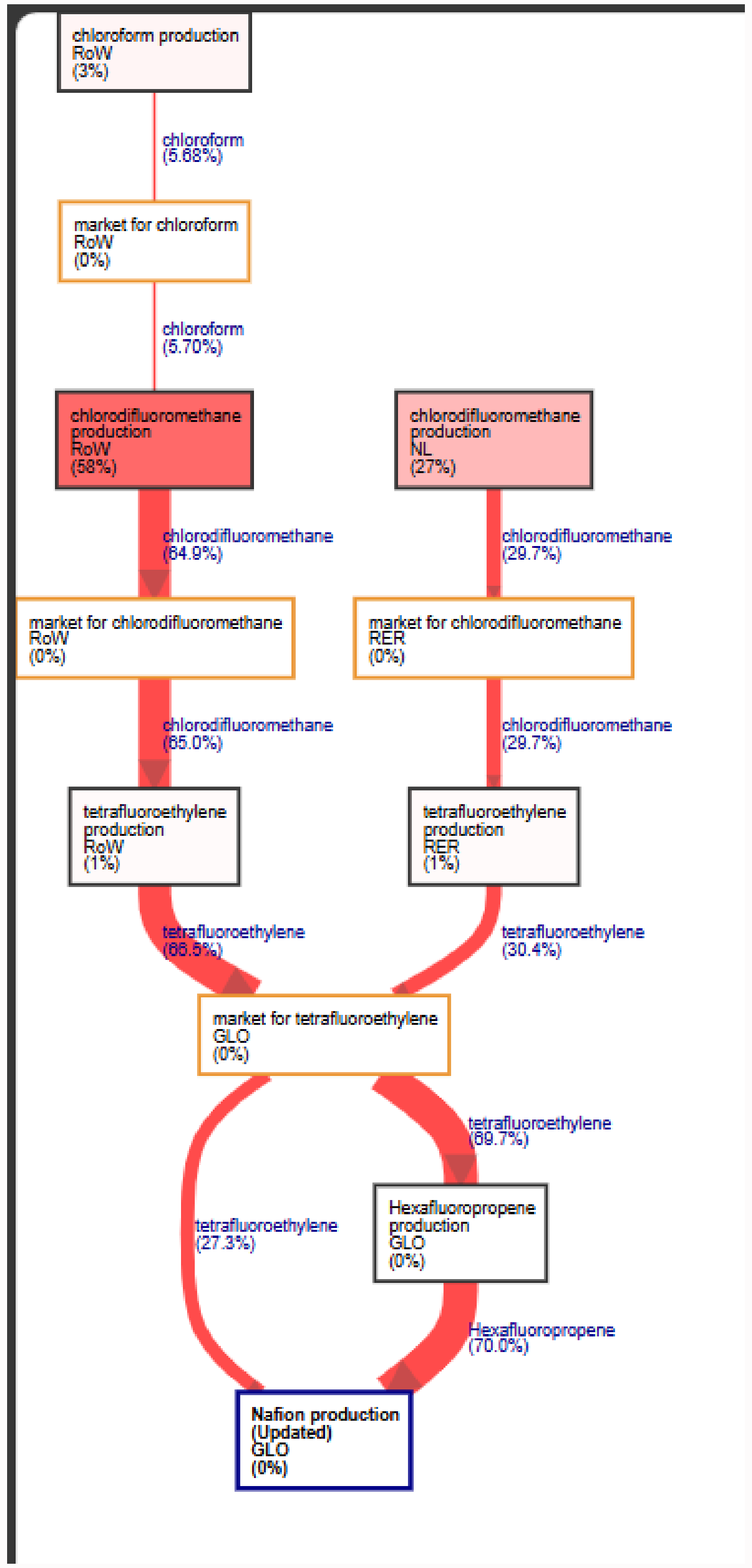


Figure 14: Virgilio et al.(2023)

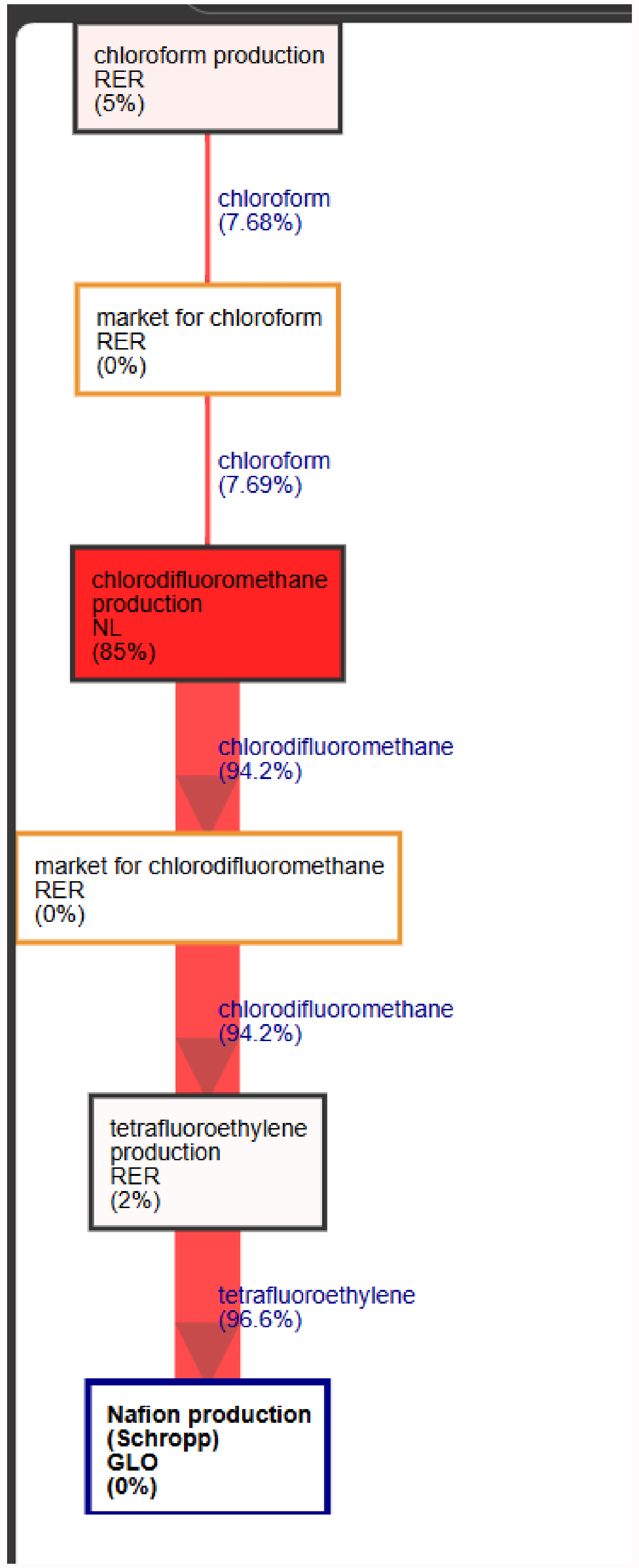


Figure 15: Schropp et al.(2024)

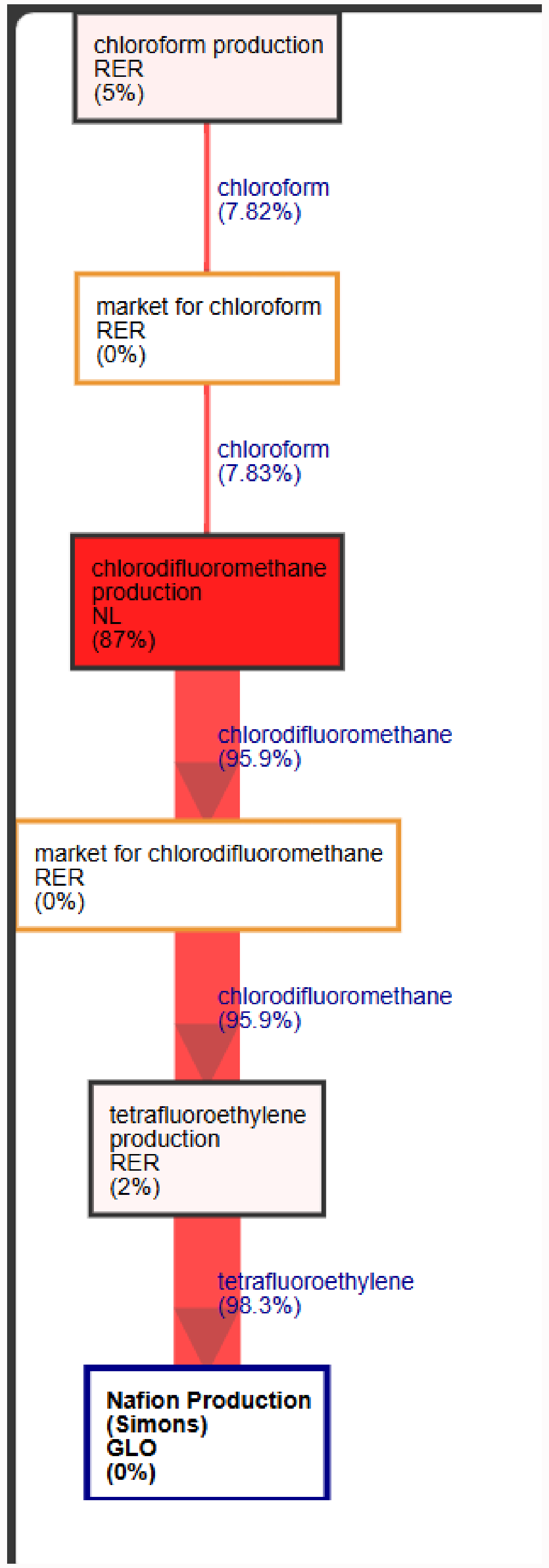


Figure 16: Simons et al.(2015)

Reference flow: 1kg Nafion production

System
Impact Contribution Analysis

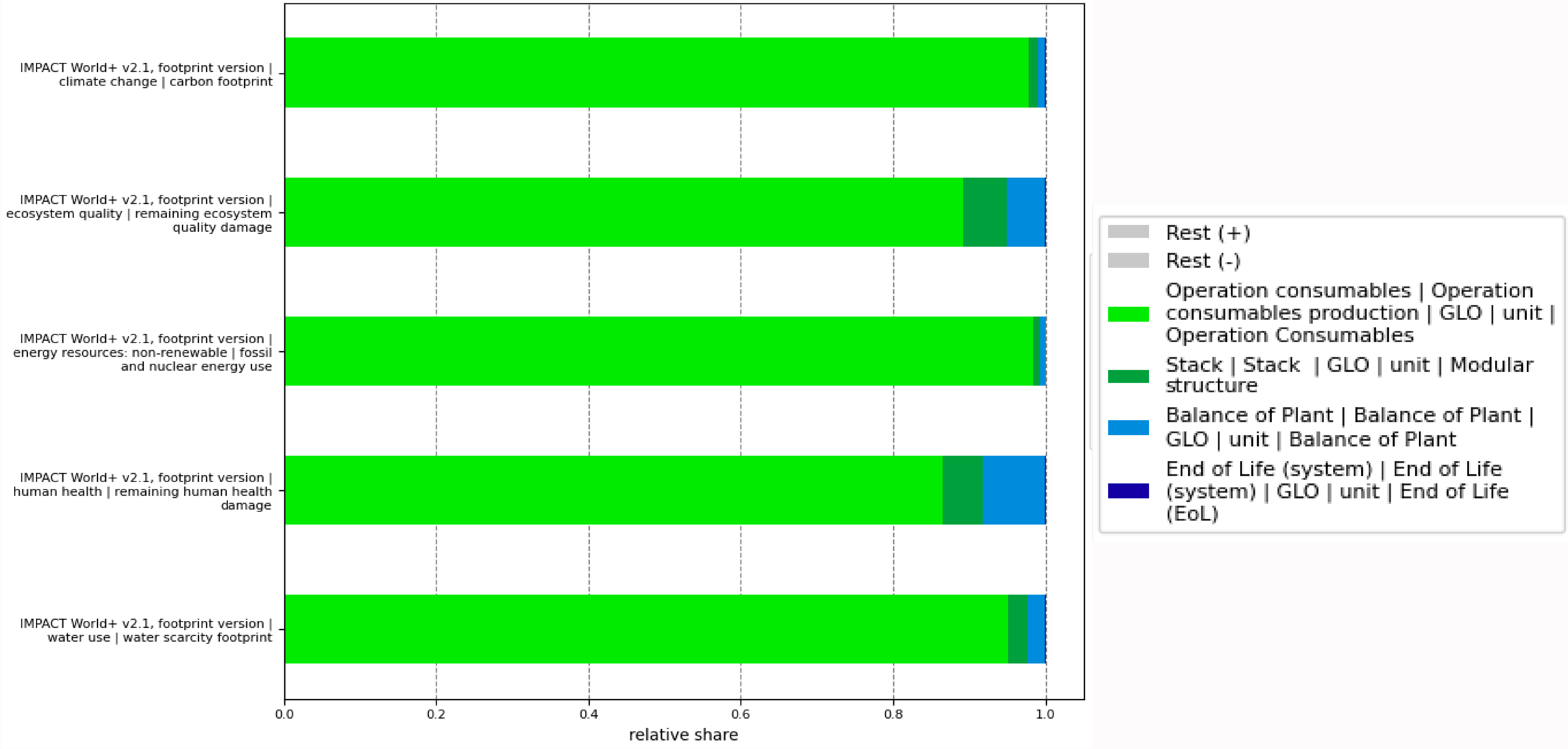


Figure 17: Reference flow: 1 unit PEMFC System

System
Impact Contribution Analysis (Processes)

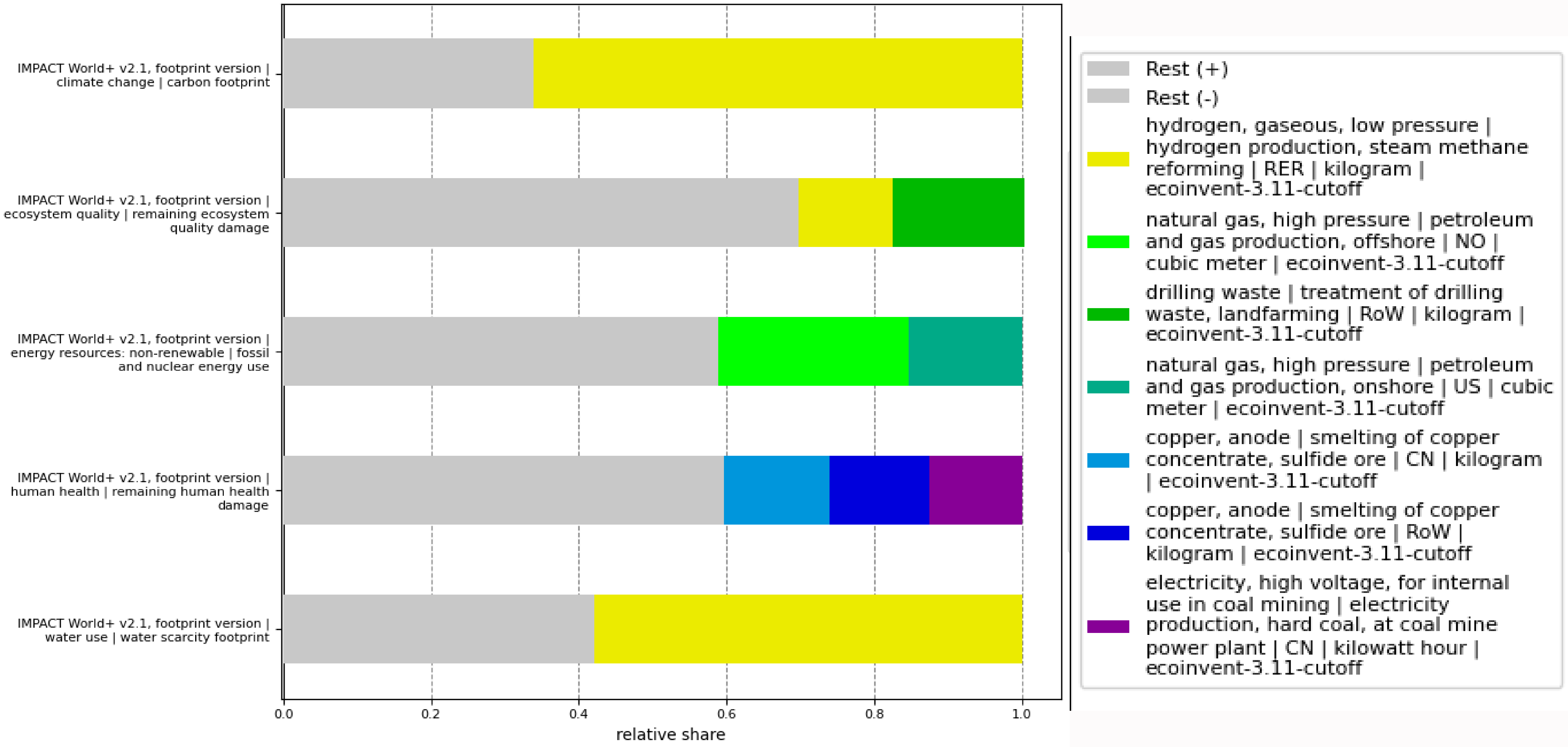


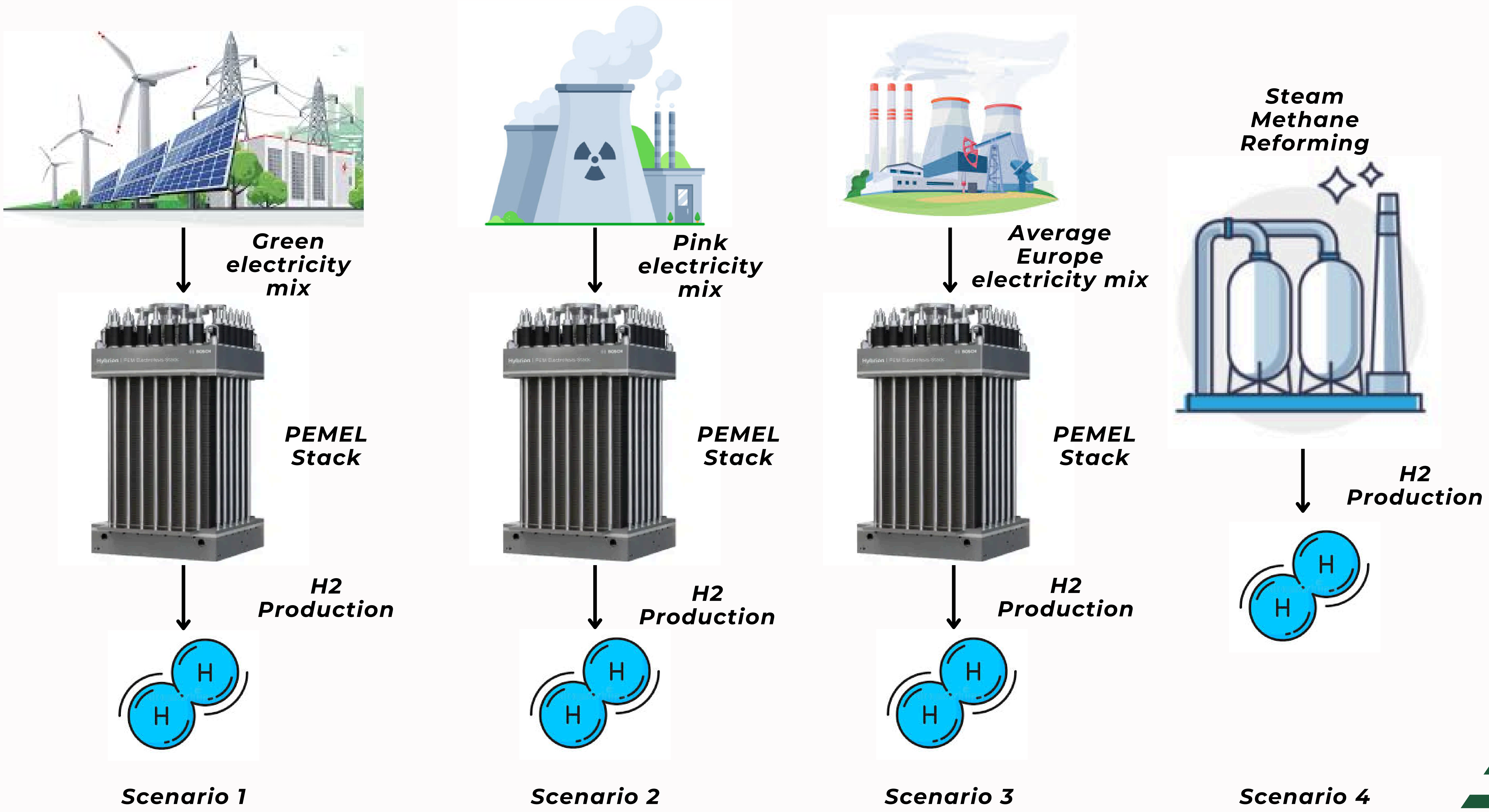
Figure 18: Reference flow: 1 unit PEMFC System

System Absolute Impact

index	climate change | carbon footprint	ecosystem quality | remaining ecosystem quality damage	energy resources: non-renewable | fossil and nuclear energy use	human health | remaining human health damage	water use | water scarcity footprint
Unit	kg CO2-eq	pdf.m^2.year	MJ deprived	DALYs	m^3 world eq deprived
Score	0.701729284	0.062476795	12.76238356	2.18E-07	0.116937039
Operation consumables	0.685914082	0.055717233	12.55128015	1.88E-07	0.11115629
Balance of Plant	0.007038209	0.003114648	0.087273037	1.78E-08	0.002790909
Stack	0.008762212	0.003640877	0.123702091	1.17E-08	0.002985162
End of Life (system)	1.48E-05	4.04E-06	0.000128285	7.69E-12	4.68E-06
H2 tool Score	0.802	0.104	19.7	3.59E-07	0.494
Score difference (%)	-12.50	-39.93	-35.22	-39.37	-76.33

Figure 19: Impact of a 100kW PEMFC system to deliver the FU, 1kWh of net electricity compared with corresponding score calculatd by H2Tool

H2 production comparison



H2 production comparison

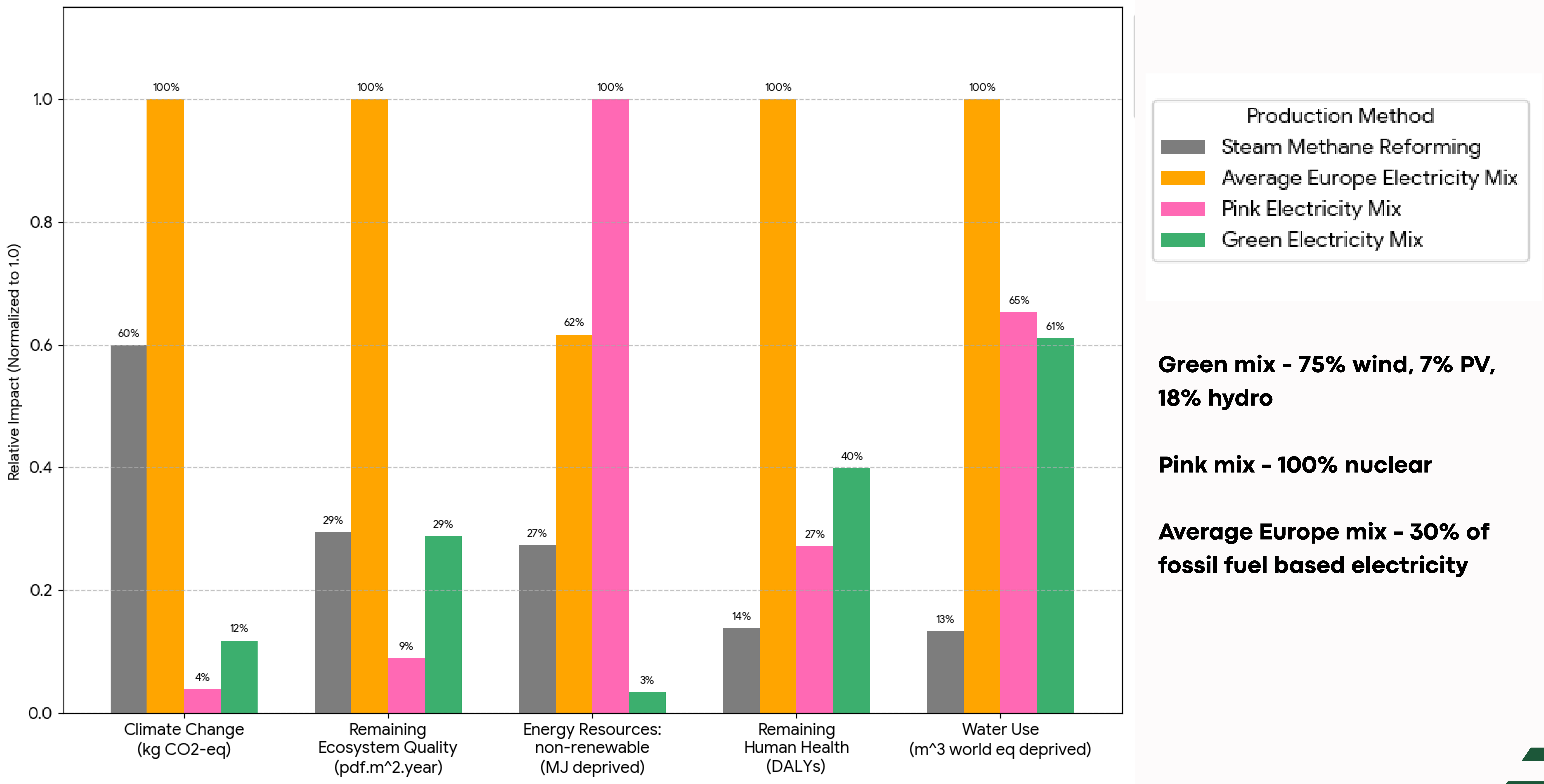


Figure 20: based on ImpactWorld+ Footprint Impact categories

Green mix - 75% wind, 7% PV, 18% hydro

Pink mix - 100% nuclear

Average Europe mix - 30% of fossil fuel based electricity

Data Source: HTWOL/H2Tool

H2 production comparison

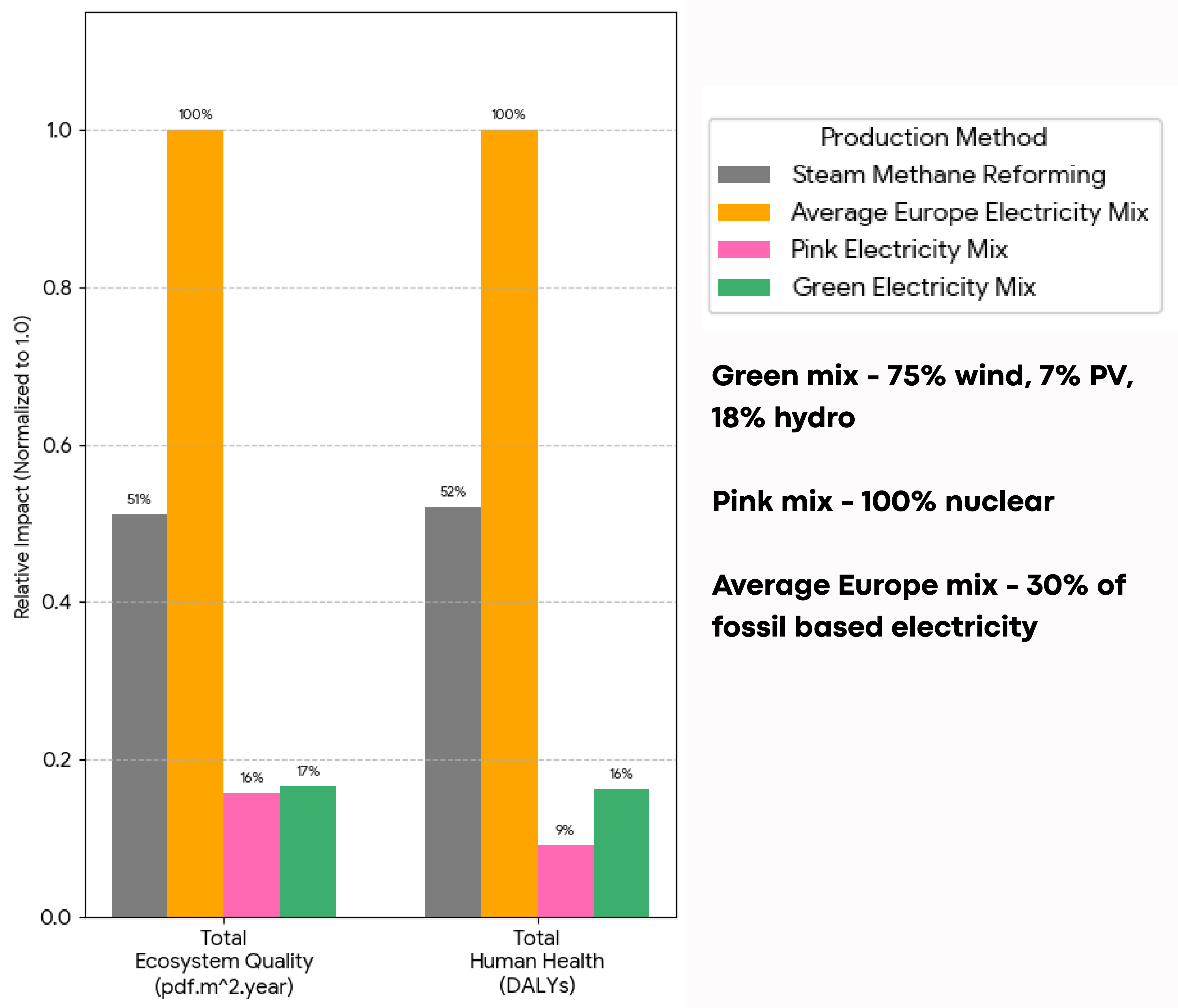


Figure 21: based on ImpactWorld+ Expert Impact categories

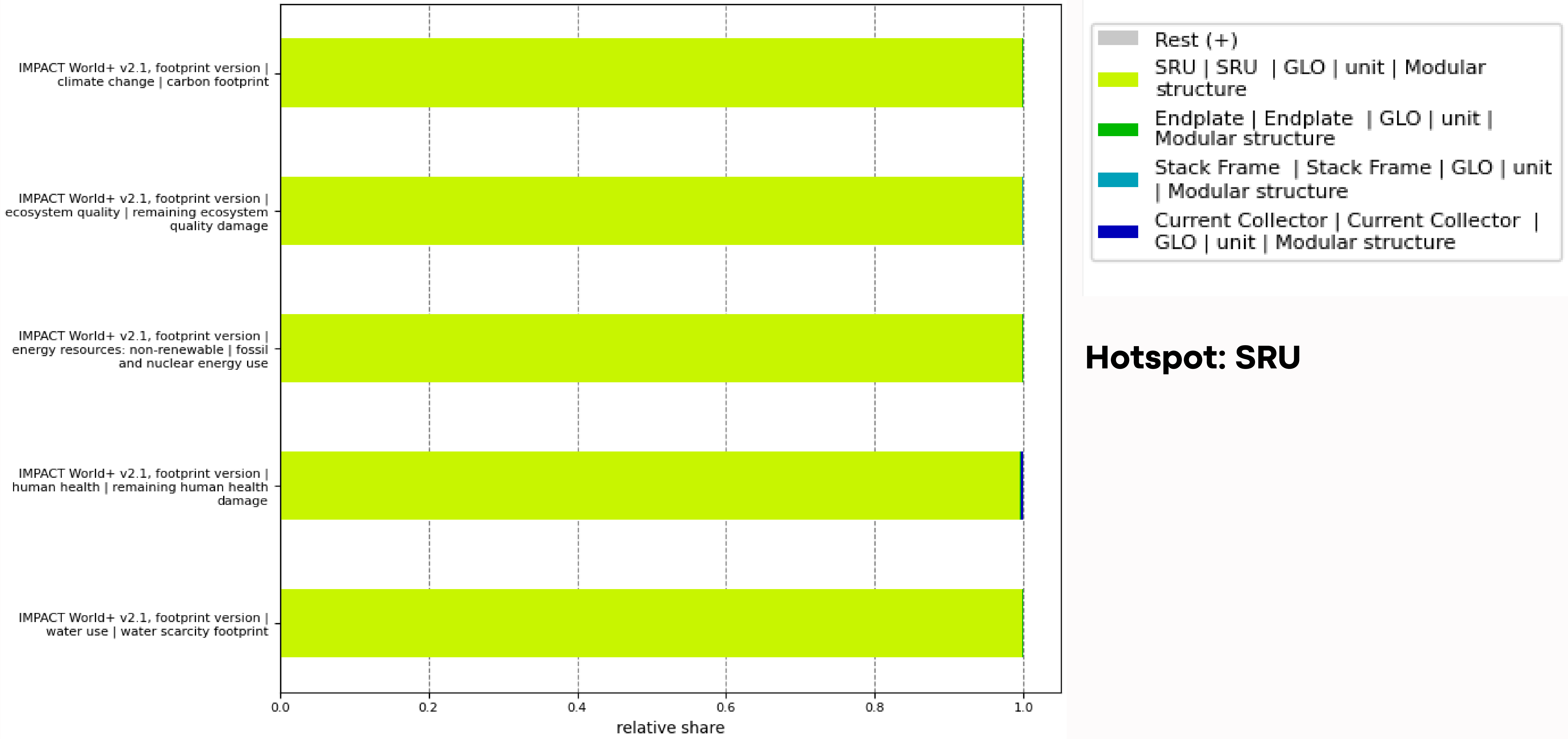
Green mix - 75% wind, 7% PV, 18% hydro

Pink mix - 100% nuclear

Average Europe mix - 30% of fossil based electricity

Data Source: HTWOL/H2Tool

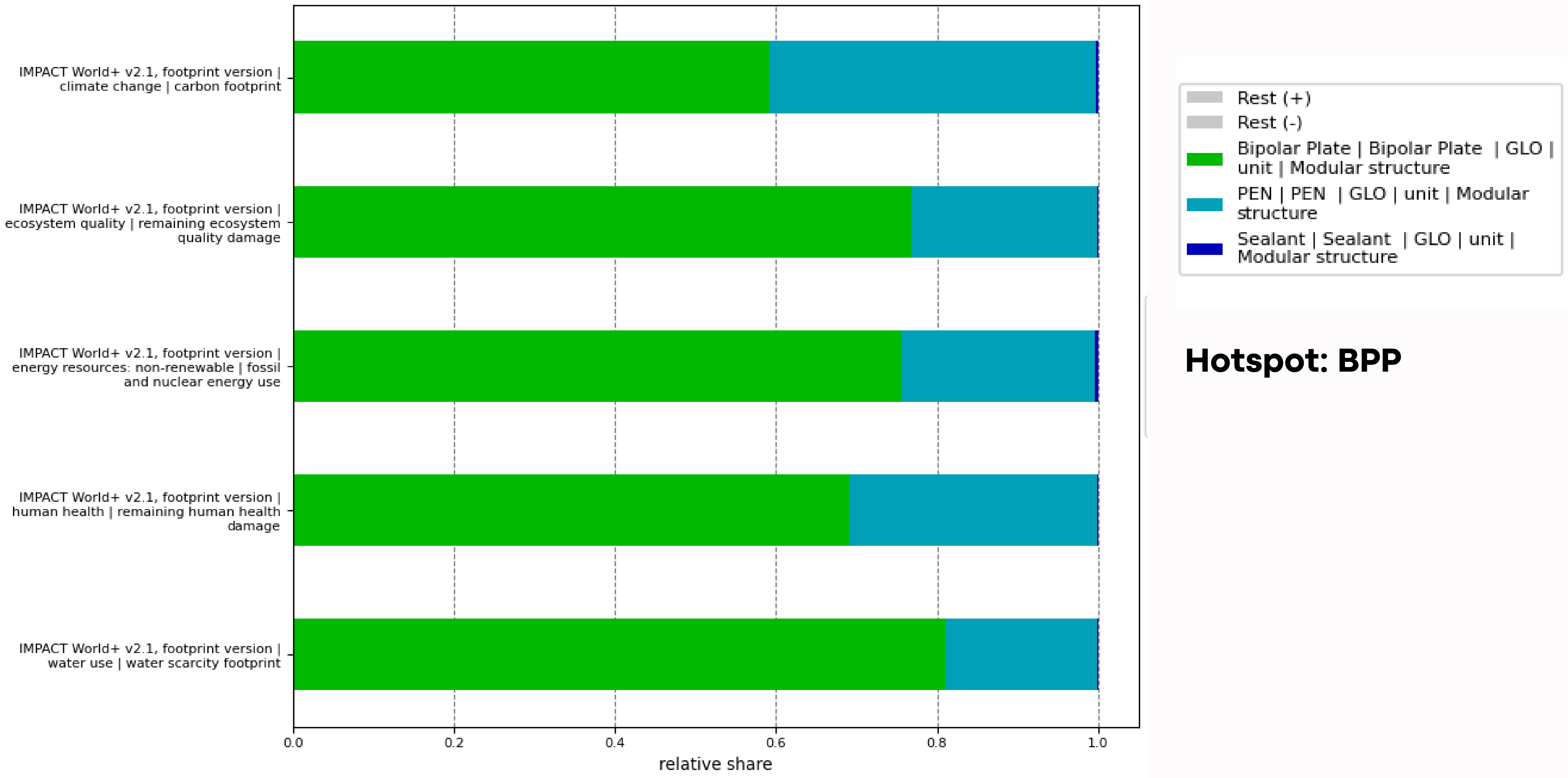
Stack (Stack Tier-2) Impact Contributions



Hotspot: SRU

Figure 22: Reference flow: 1 unit PEMFC Stack

SRU (Stack Tier-3) Impact Contributions



Hotspot: BPP

Figure 23: Reference flow: 1 unit SRU

Riemer et al.(2023)
Stack Contributions

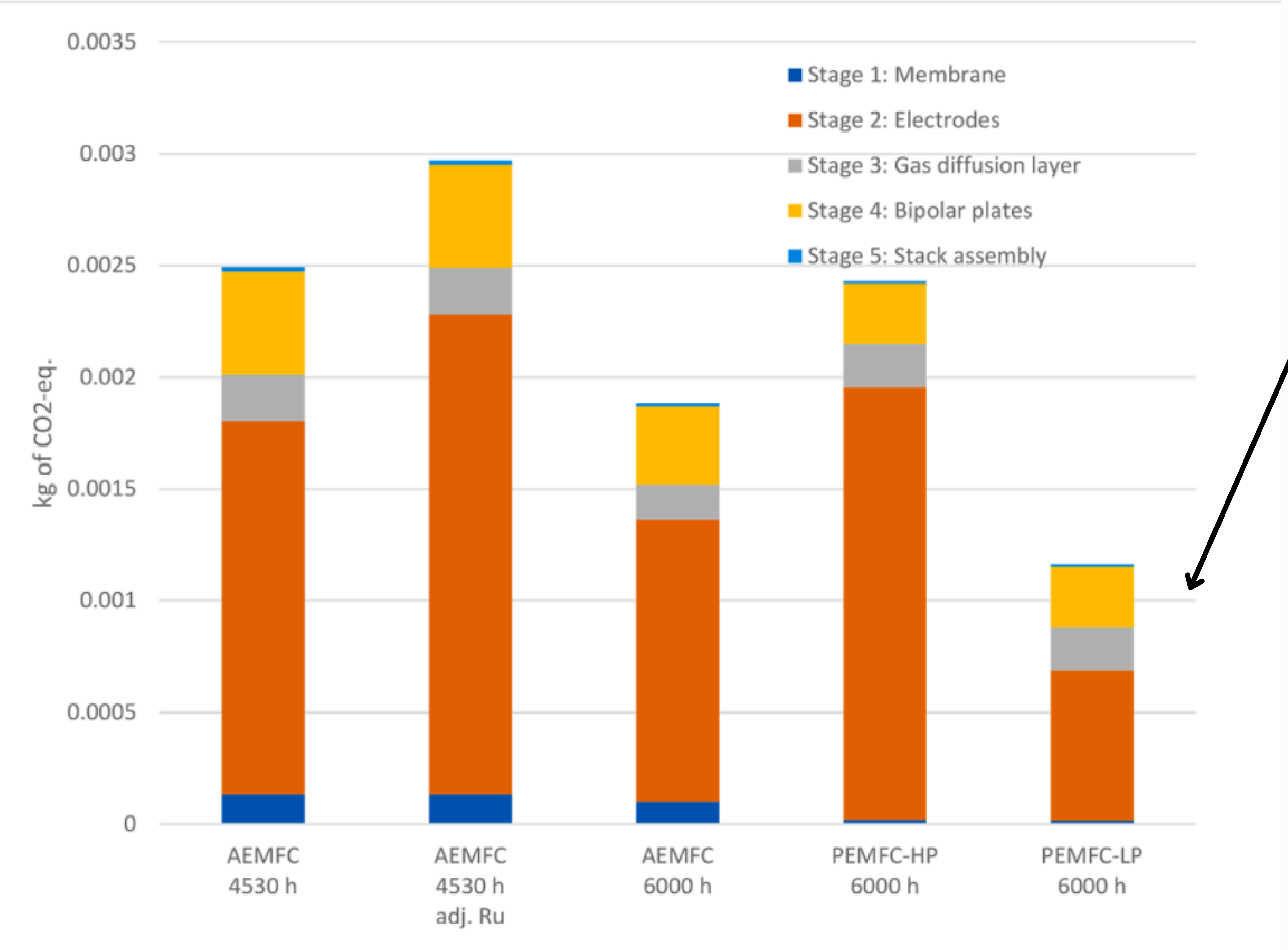


Figure 24: Reference flow: 1 unit 100kW PEMFC Stack

This data here is slightly different from my LCA results since the contribution of PEN dominates that of BPP.

BPP Contributions
Climate change, Carbon footprint

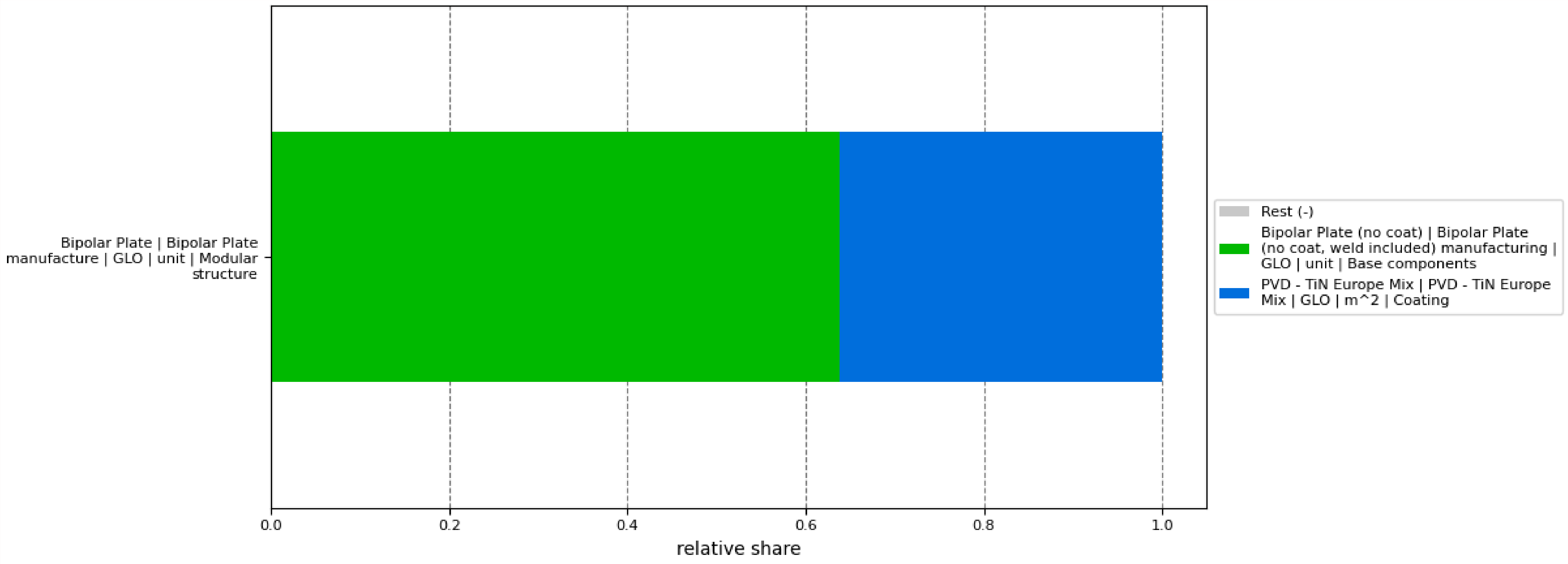
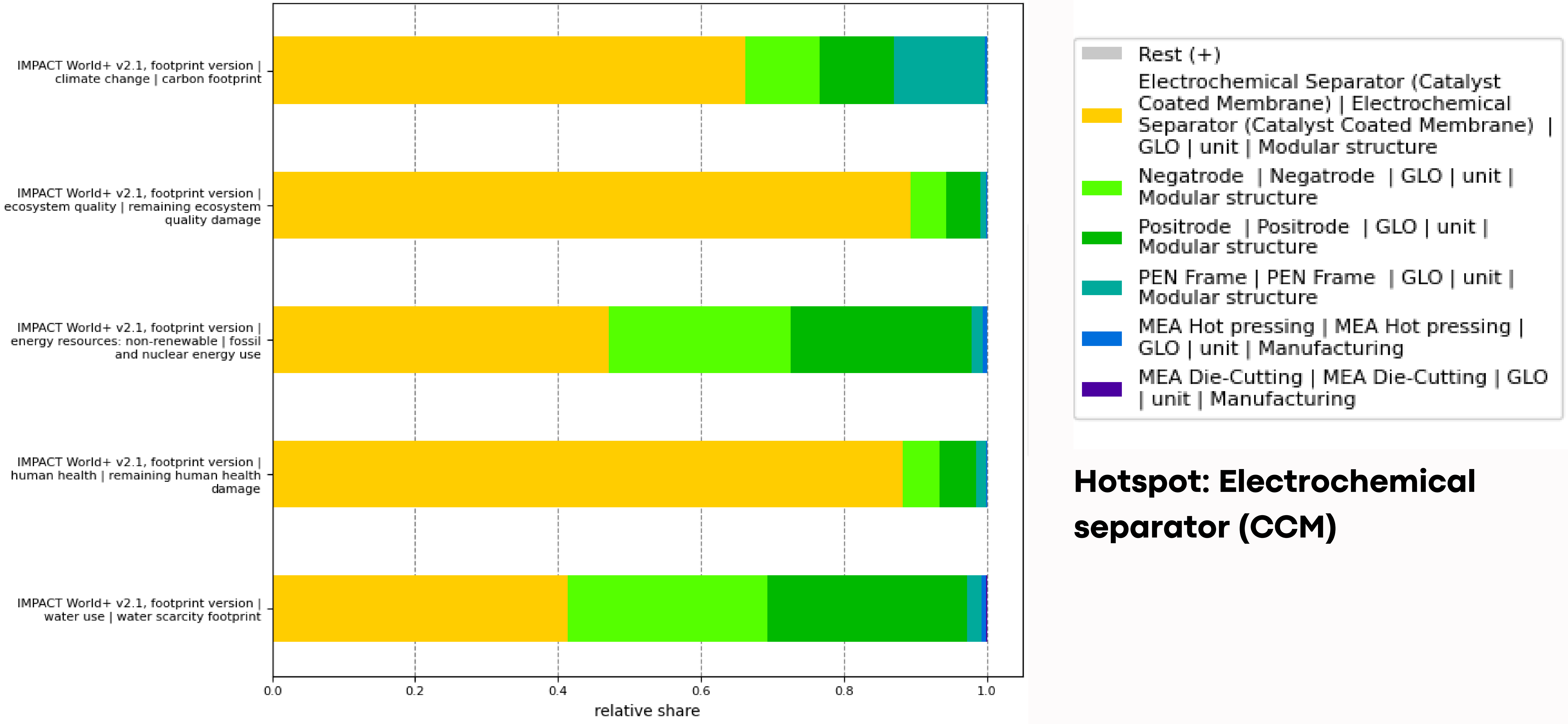


Figure 25: Reference flow: 1 unit BPP

This due to the state-of-the-art PVD based TiN coating included in my model, as opposed to no coating in Riemer's model.

PEN (Stack Tier-4) Impact Contributions



Hotspot: Electrochemical separator (CCM)

Figure 26: Reference flow: 1 unit PEN

Electrochemical Separator (CCM) (Stack Tier-5) Impact Contributions

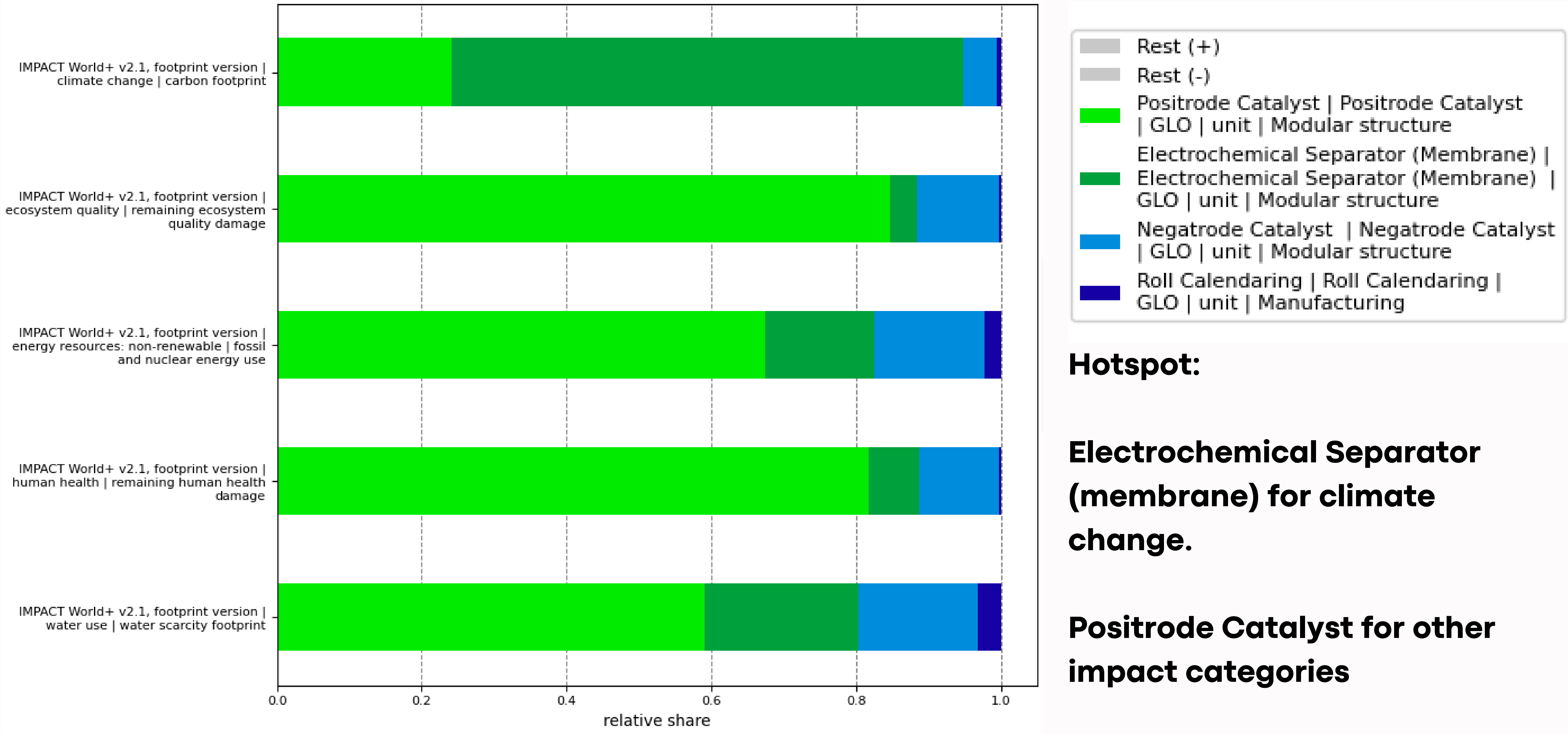


Figure 27: Reference flow: 1 unit CCM

Hotspot:

Electrochemical Separator (membrane) for climate change.

Positrode Catalyst for other impact categories

Electrochemical Separator (CCM) (Stack Tier-5) Impact Contributions (Processes)

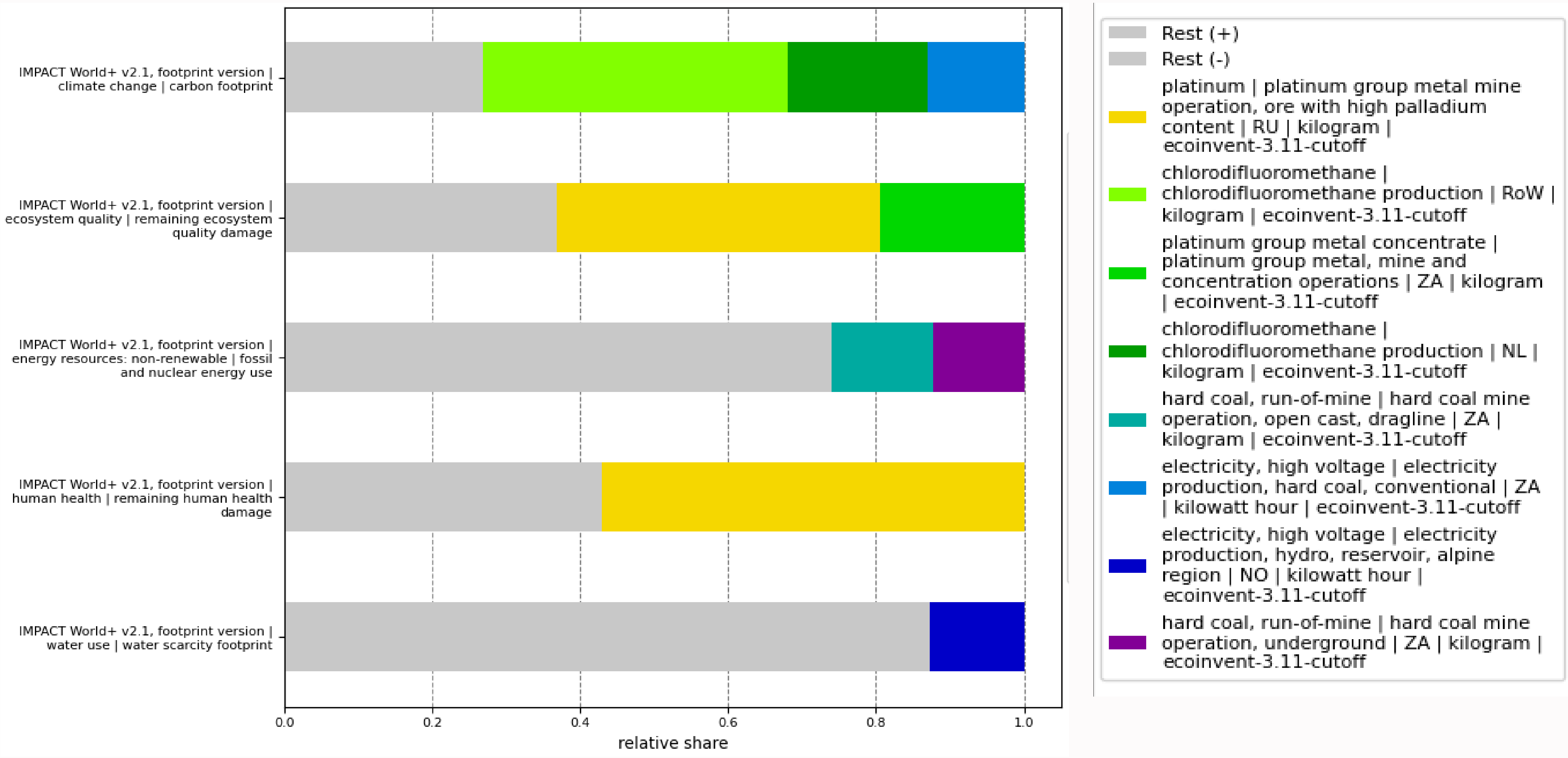


Figure 28: Reference flow: 1 unit CCM

- Production of chlorodifluoromethane, which is used to produce terafluoroethylene, an essential component for Nafion (membrane) production is the major contributor in climate change category.
- Process related to production of contribute majorly in the Remaining Ecosystem Quality and Remaining Human Health Impact categories.

Sensitivity Analysis of Dynamic Parameters

1. Current density
2. Cell voltage
3. Stack lifetime
4. Surplus hydrogen
5. BoP lifetime

Sensitivity Analysis of Dynamic Parameters									Impact				
+/- 10% of original value is used for SA									CC	Rem. EQ	ER	Rem HH	Water use
Parameter	**Magnitude**	**Unit**	**Power**	**Unit**	**Operation time**	**Unit**	**Lifetime**	**Unit**	kg CO2-eq	pdf.m^2.year	MJ deprived	DALYs	m^3 world eq deprived
Current density	1	A/cm^2	100	kW	36	sec	6000	hrs	0.70172928	0.06247680	12.762384	0.00000022	0.11693704
Current density (-10%)	0.9	A/cm^3	89.99892	kW	40.00048001	sec	6000	hrs	0.77421107	0.067428575	14.085032	2.28E-07	0.12734444
Current density (+10%)	1.1	A/cm^4	109.99868	kW	32.72766546	sec	6000	hrs	0.64244027	0.058426314	11.680477	2.09E-07	0.10842394
Cell voltage	0.68	V	100	kW	36	sec	6000	hrs	0.70172928	0.06247680	12.762384	0.00000022	0.11693704
Cell voltage (-10%)	0.612	V	89.93952	kW	40.02689808	sec	6000	hrs	0.77468976	0.067461277	14.093767	2.28E-07	0.12741317
Cell voltage (+10%)	0.748	V	109.92608	kW	32.74928024	sec	6000	hrs	0.64283186	0.058453067	11.687623	2.10E-07	0.10848017
Stack Lifetime	6000	hrs	100	kW	36	sec	6000	hrs	0.70172928	0.06247680	12.762384	0.00000022	0.11693704
Stack Lifetime (-10%)	5400	hrs	100	kW	36	sec	5400	hrs	0.7027045	0.062881786	12.776142	2.19E-07	0.11726924
Stack Lifetime (+10%)	6600	hrs	100	kW	36	sec	6600	hrs	0.70093138	0.06214544	12.751126	2.17E-07	0.11666524
Surplus Hydrogen	0.084785	kgH2/hr	100	kW	36	sec	6000	hrs	0.70172928	0.06247680	12.762384	0.00000022	0.11693704
Surplus Hydrogen (-10%)	0.0763065	kgH2/hr	100	kW	36	sec	6000	hrs	0.70075787	0.062419099	12.744539	2.18E-07	0.11680291
Surplus Hydrogen (+10%)	0.0932635	kgH2/hr	100	kW	36	sec	6000	hrs	0.70270074	0.062534495	12.780229	2.18E-07	0.11707117
BoP Lifetime	80000	hrs	100	kW	36	sec	6000	hrs	0.70172928	0.06247680	12.762384	0.00000022	0.11693704
BoP Lifetime	72000	hrs	100	kW	36	sec	6000	hrs	0.70251131	0.062822867	12.772081	2.20E-07	0.11724714
BoP Lifetime	88000	hrs	100	kW	36	sec	6000	hrs	0.70108945	0.062193646	12.754450	2.16E-07	0.11668332

Figure 29: One at a time SA of dynamic parameters

Dynamic Parameters SA
Climate Change

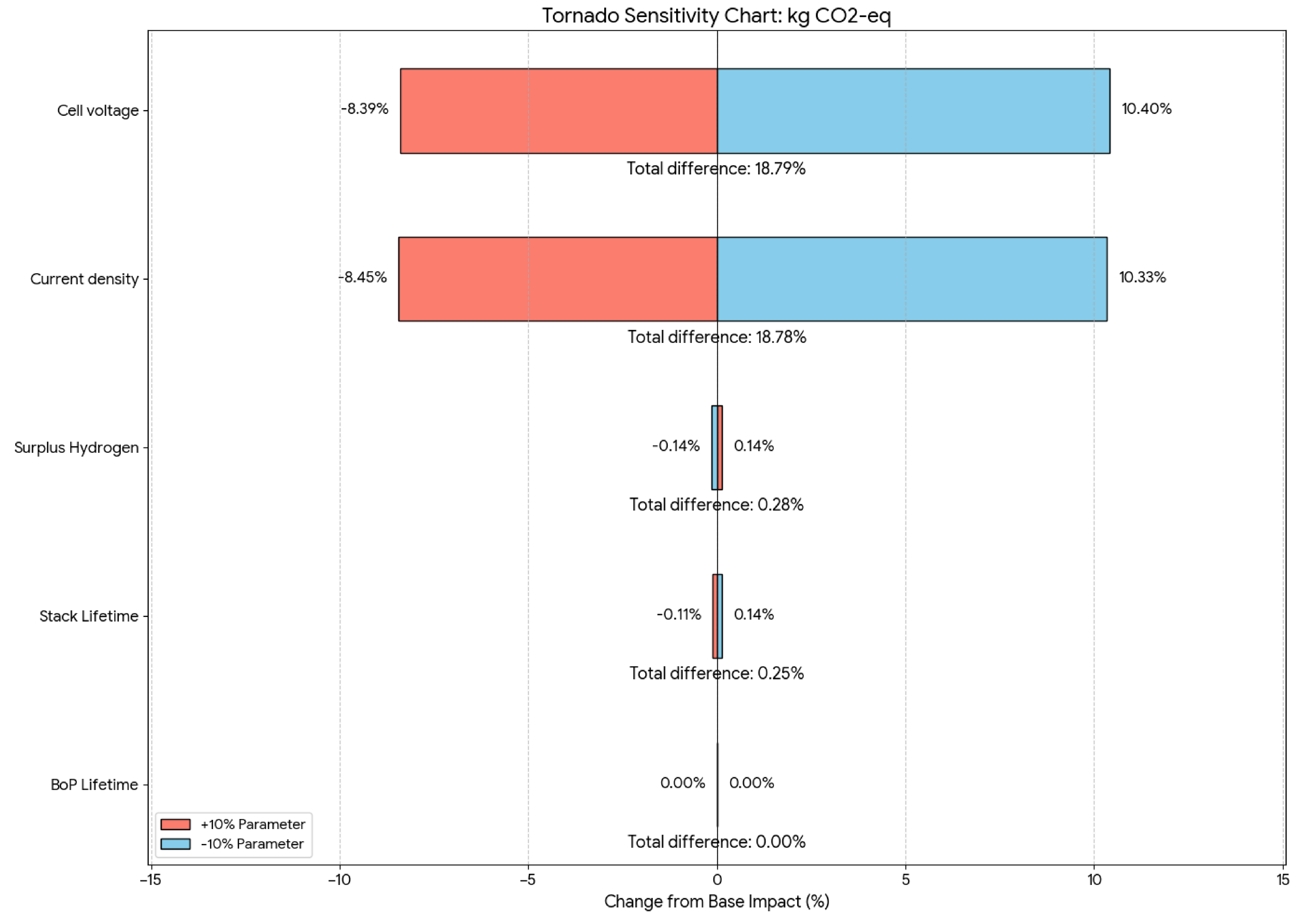


Figure 30

Dynamic Parameters SA
Remaining Ecosystem Quality

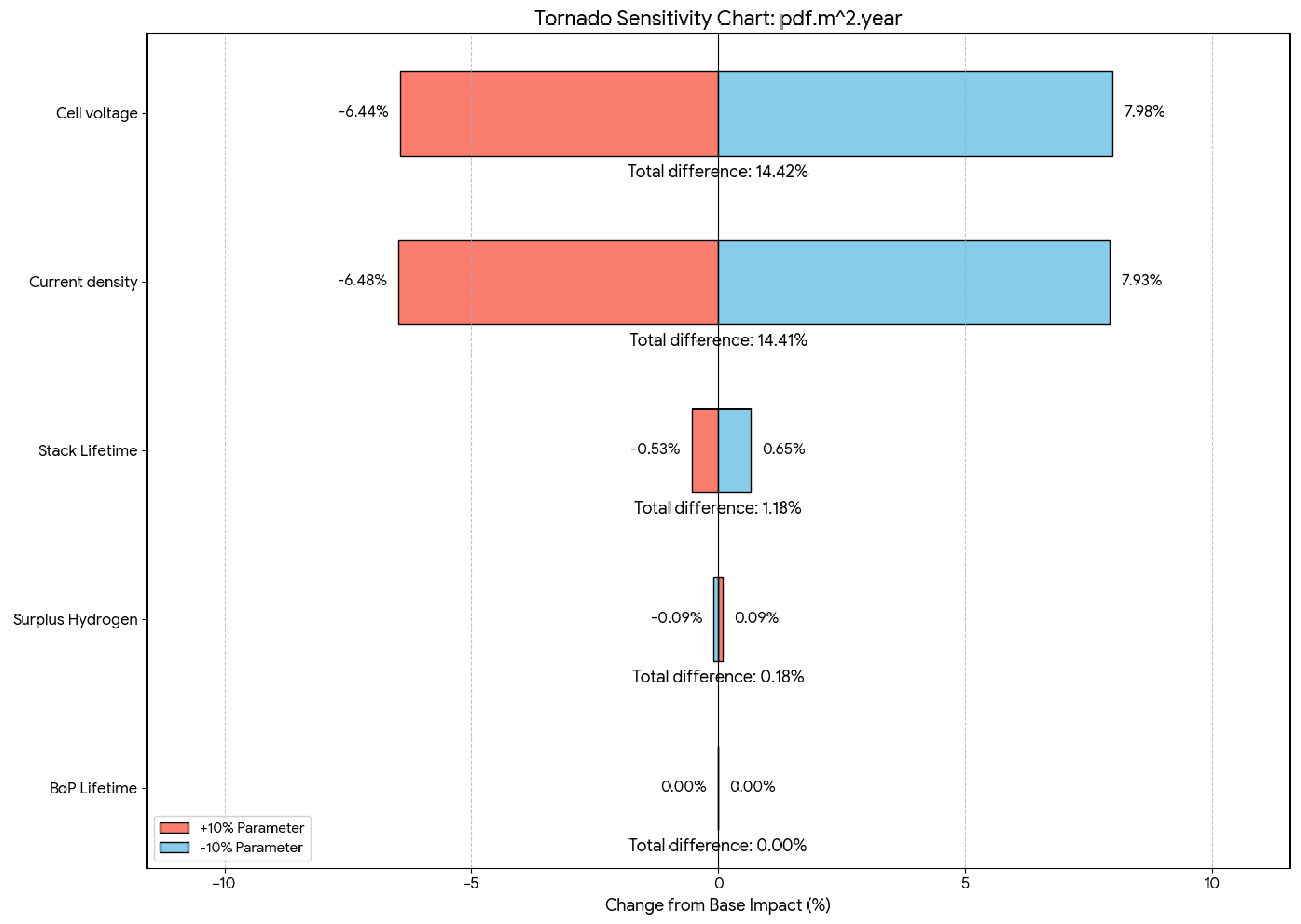


Figure 31

Dynamic Parameters SA
Energy Resources: Non-renewable

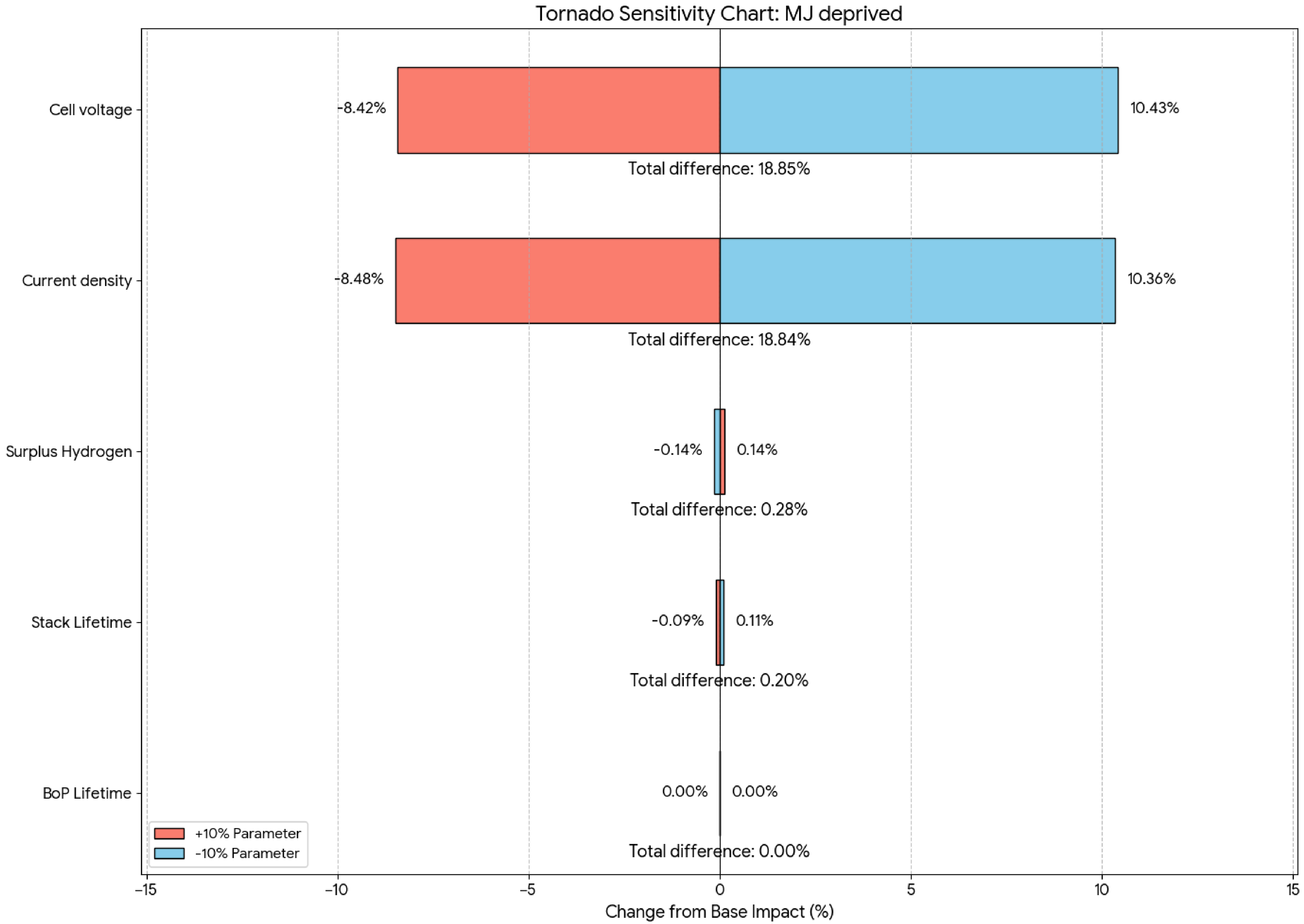


Figure 32

Dynamic Parameters SA
Remaining Human Health

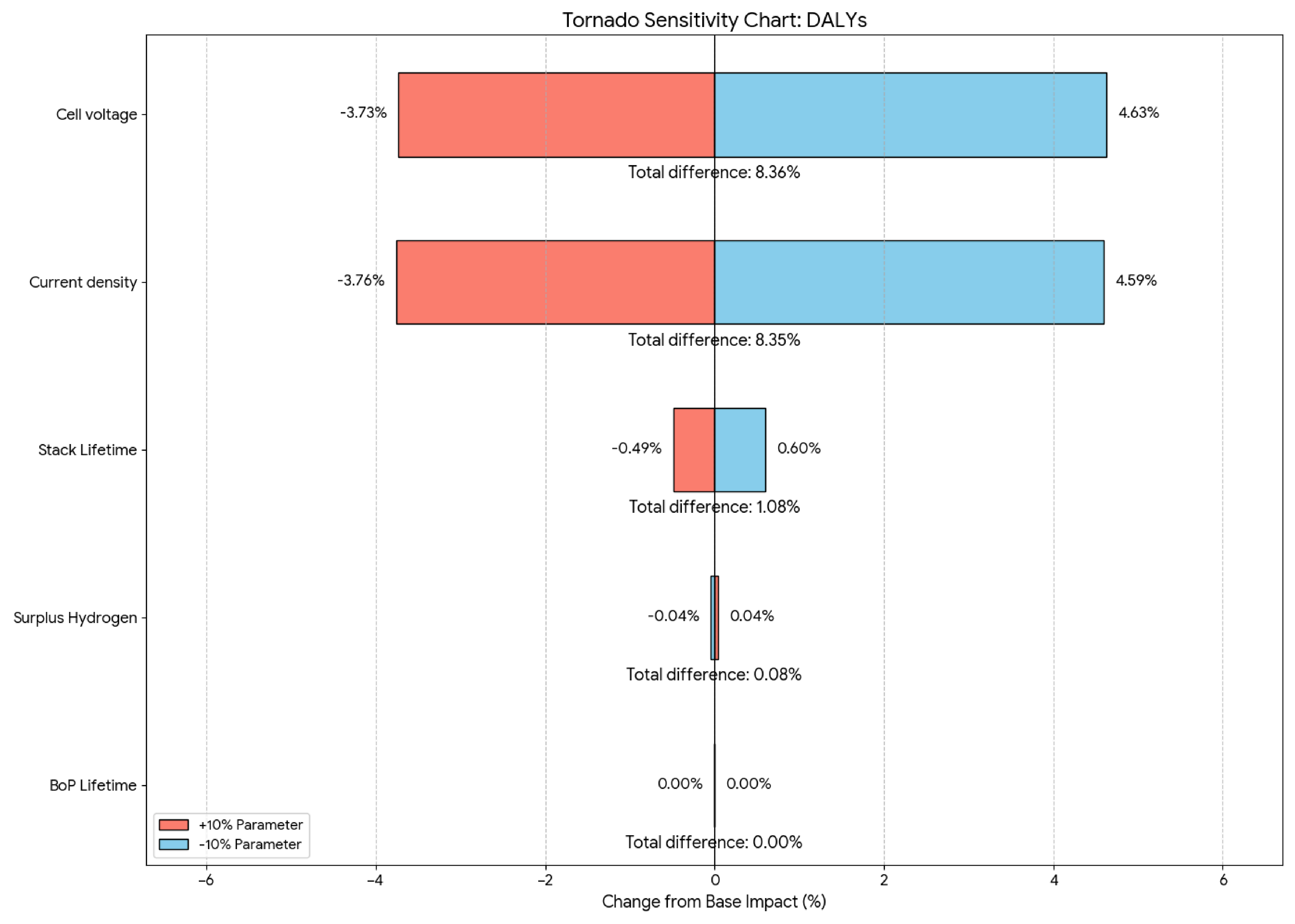


Figure 33

Dynamic Parameters SA
Water Use

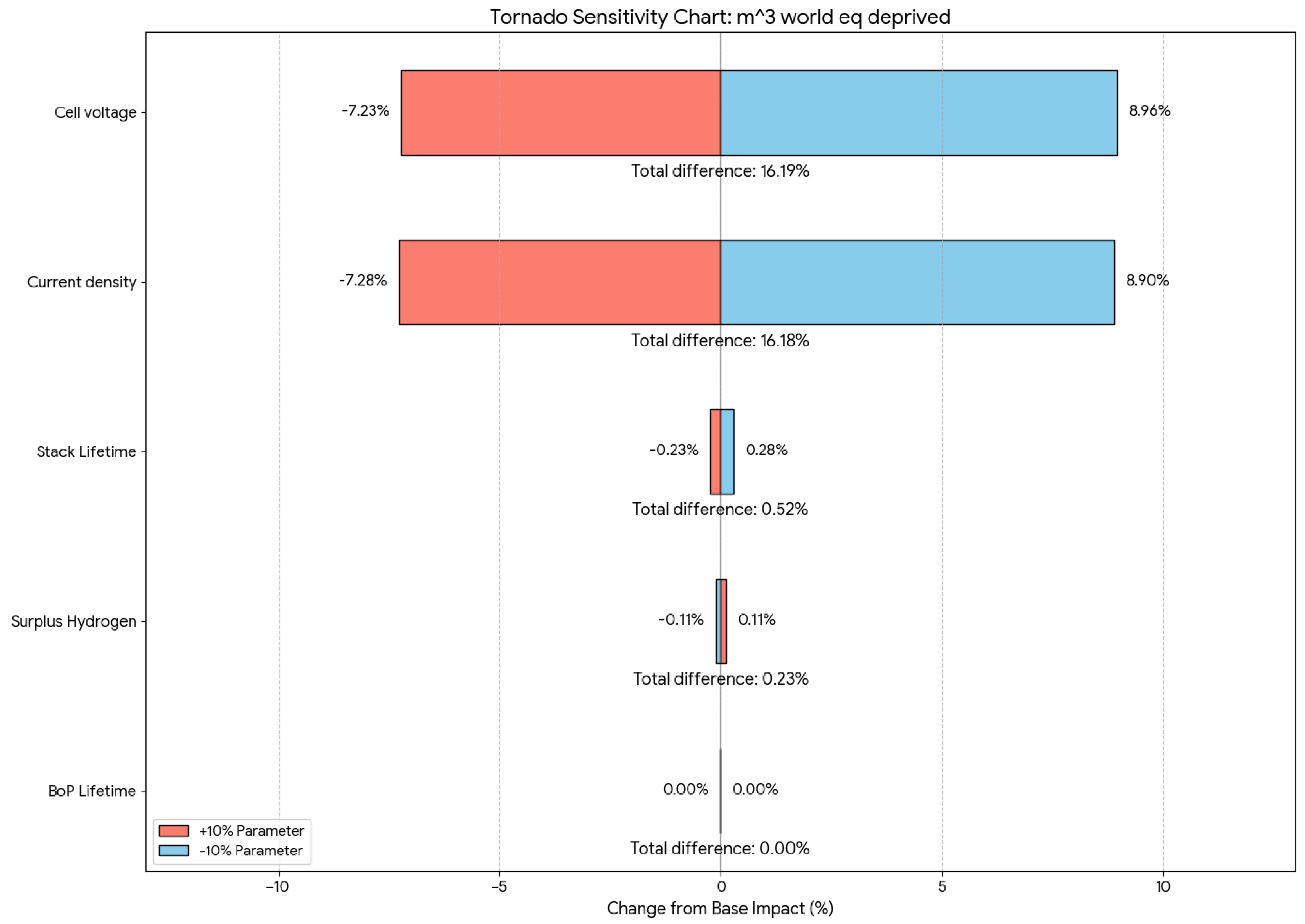


Figure 34

Thank You